Weak convergence and empirical processes

Soumendu Sundar Mukherjee
Indian Statistical Institute, Kolkata

April 24, 2019

Warning: These course notes (for M.Stat. second year students) have not been subject to very careful scrutiny, and so there may be various typos/mistakes here and there. Please let me know at soumendu041@gmail.com if you find one.

Contents

1 Review of metric topology **3**
1.1 Metric spaces . . . 3
1.2 Open and closed sets . . . 4
1.3 Continuity . . . 4
1.4 Limit points and accumulation points . . . 5
1.5 Denseness . . . 6
1.6 Convergence, Cauchy sequences and completeness . . . 6
1.7 Compactness . . . 7
1.8 Connectedness . . . 8

2 Probability measures on metric Spaces **8**
2.1 Preliminaries . . . 8
2.2 Weak convergence . . . 9
2.3 The mapping theorem and Skorohod's representation . . . 11
2.4 Relative compactness and Prohorov's theorem . . . 12

3 Weak convergence in $C[0,1]$ **13**
3.1 Tightness in $C[0,1]$ . . . 13
3.2 Wiener measure and Donsker's theorem . . . 13
3.3 Some consequences of Donsker's theorem . . . 16
3.4 The (standard) Brownian bridge . . . 17
3.5 Brownian motion on $[0,\infty)$ . . . 18

4 Weak convergence in $D[0,1]$ **20**
4.1 The Skorohod topology . . . 20

5 More on Empirical Processes **22**
5.1 Concentration inequalities 23
5.2 Covering number and bracketing numbers 25
5.3 Symmetrization 26
5.4 Discretization and chaining 28

1 Review of metric topology

We will do a very condensed review of metric space topology here. There will be numerous exercises which you should attempt to do on your own. Having a good grasp of the concepts discussed in this section is *necessary* for this course. You may consult Kumaresan (2005) and Munkres (1975) for more details.

1.1 Metric spaces

Let $\mathcal{M}$ be a non-empty set. $\rho : \mathcal{M} \times \mathcal{M} \to \mathbb{R}_+ (= [0, \infty))$ is called a *metric* if it satisfies

1. $\rho(x, y) = \rho(y, x) \forall x, y \in \mathcal{M}$.
2. $\rho(x, y) = 0$ if and only if $x = y$.
3. $\rho(x, y) \leq \rho(x, z) + \rho(z, y) \forall x, y, z \in \mathcal{M}$.

Together the pair $(\mathcal{M}, \rho)$ is called a *metric space*.

Example 1.1. *$\mathcal{M} = \mathbb{R}^d, \rho = \ell_p(x, y) = (\sum_{i=1}^d |x_i - y_i|^p)^{1/p}$, with $1 \leq p < \infty$. For $p = \infty$, $\ell_p(x, y) = \max_{1 \leq i \leq d} |x_i - y_i|$. Important special cases are the ℓ_1, ℓ_2, and the ℓ_∞ metrics. ℓ_2 is the standard Euclidean metric on $\mathbb{R}^d$.*

Exercise 1.2. *Show that $\ell_p(x, y) \overset{p \to \infty}{\to} \ell_\infty(x, y)$, for all $x, y \in \mathbb{R}^d$.*

Example 1.3. *$\mathcal{M}$ is any non-empty set. Let*

$$\rho(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{otherwise.} \end{cases}$$

Then ρ is called the discrete metric, and $(\mathcal{M}, \rho)$ is called a discrete metric space.

Example 1.4. *Let $\mathcal{M} = C[0, 1]$, the space of all real-valued continuous functions on $[0, 1]$. $\mathcal{M}$ becomes a metric space with the metric*

$$\rho(f, g) = \left(\int_0^1 |f(t) - g(t)|^p \, dt \right)^{\frac{1}{p}},$$

$1 \leq p < \infty$. For $p = \infty$, we get the uniform metric

$$\rho(f, g) = \max_{t \in [0,1]} |f(t) - g(t)|.$$

$C[0, 1]$ with the uniform metric will a central object in this course. We will be interested in various topological properties of this metric space.

Example 1.5. *Let $\mathcal{M} = (V, E)$ be a finite graph and define*

$$\rho(v_1, v_2) = \text{length of the shortest path joining } v_1 \text{ and } v_2.$$

If no such path exists, we define $\rho(v_1, v_2) = +\infty$ in accordance with the convention $\inf \emptyset = +\infty$.

More structure can be given beyond just a metric when $\mathcal{M}$ is also a linear space (vector space). Such as norms, or inner products.

1.2 Open and closed sets

Let $x \in \mathcal{M}$ and $\epsilon > 0$. Then

- $B(x;\epsilon) := \{y \in \mathcal{M} \mid \rho(x,y) < \epsilon\}$ is the open ball of radius ϵ around x.
- $B[x;\epsilon] := \{y \in \mathcal{M} \mid \rho(x,y) \leq \epsilon\}$ is the closed ball of radius ϵ around x.
- $G \subset \mathcal{M}$ is called *open* if for every point $x \in U$, there is an open ball $B(x;\epsilon_x) \subset U$.
- $F \subset \mathcal{M}$ is called *closed* if F^c is open.
- An open subset U containing $x \in \mathcal{M}$ is called an (open) *neighbourhood* of x.

How do open and closed sets behave under union and intersections?

- Arbitrary union (resp. intersection) of open (resp. closed) sets is open (resp. closed).
- Finite intersection (resp. union) of open (resp. closed) sets is open (resp. closed).

Let $A \subset \mathcal{M}$. The *closure* of A, denoted by $\bar{A}$, is the set $\cap_{F \supset A, F \text{ closed}} F$, i.e. $\bar{A}$ is the "smallest" closed set containing A. The *interior* of A, denoted by A°, is the set $\cup_{G \subset A, F \text{ open}} G$, i.e. A° is the "largest" open set containing A. The boundary of A, denoted by ∂A, is the set $\bar{A} \setminus A^\circ$.

Exercise 1.6. *Prove or disprove (a) $\overline{B(x;\epsilon)} = B[x;\epsilon]$, (b) $B[x;\epsilon]^\circ = B(x;\epsilon)$, and (c) $\partial B[x;\epsilon] = \partial B(x;\epsilon) = S(x;\epsilon) = \{y \in \mathcal{M} \mid \rho(y,x) = \epsilon\}$.*

Together, the collection of all open sets of $\mathcal{M}$ defines its *topology*.

Side remark: Arbitrary intersection (resp. union) of open (resp. closed) sets are called G_δ (resp. F_σ) sets. What about arbitrary union/intersection of G_δ or F_σ sets? One can actually build a hierarchy of sets in this way: $G_{\delta\sigma}, G_{\delta\sigma\delta}, \ldots, F_{\sigma\delta}, F_{\sigma\delta\sigma}, \ldots$ and so on.

Exercise 1.7. *Show that every open (resp. closed) set is F_σ (resp. G_δ).*

Exercise 1.8. *What are all the open (closed) sets in a discrete metric space?*

1.3 Continuity

Let $\mathcal{M}_1, \mathcal{M}_2$ be metric spaces, and $f : \mathcal{M}_1 \to \mathcal{M}_2$. We say that f is *continuous at x* if, for any neighbourhood V of $f(x)$ in $\mathcal{M}_2$, there exists a neighbourhood U of x in $\mathcal{M}_1$ such that $U \subset f^{-1}(V)$.

Exercise 1.9. *Reformulate this as the usual (ϵ, δ)-definition.*

If f is continuous at all points $x \in M_1$, then we simply say that f is continuous.

Exercise 1.10. *Show that $f : \mathcal{M}_1 \to \mathcal{M}_2$ is continuous if and only if, for all open $V \subset \mathcal{M}_2$, $f^{-1}(V)$ is open in $\mathcal{M}_1$.*

Side remark. If if, for all open $U \subset \mathcal{M}_1$, $f(U)$ is open in $\mathcal{M}_1$, then we call f an *open* map. An invertible map that is continuous as well as open is called a *homeomorphism*.

Exercise 1.11. *What are the continuous functions when $\mathcal{M}_1$ is discrete?*

A function $f : \mathcal{M}_1 \to \mathcal{M}_2$ is called *uniformly continuous*, if, for all $\epsilon > 0$, $\exists \delta > 0$ such that $\rho_1(x, y) < \delta \implies \rho_2(f(x), f(y)) < \epsilon$.

Exercise 1.12. *Of course, all uniformly continuous functions are continuous. Give an example of a function that is continuous but not uniformly continuous.*

Let $A \subset \mathcal{M}, x \in \mathcal{M}$. Define the distance of x from A as

$$\rho(x, A) := \inf_{y \in A} \rho(x, y).$$

Exercise 1.13. *Show that for any $x, y \in \mathcal{M}$,*

$$|\rho(x, A) - \rho(y, A)| \leq \rho(x, y).$$

Hence, the map $x \mapsto \rho(x, A)$ is uniformly continuous.

Lemma 1.14 (Urysohn's lemma)**.** *Let A, B be two closed subsets in $\mathcal{M}$. Then there exists a continuous function $f : \mathcal{M} \to \mathbb{R}$ such that $f(A) = 0$ and $f(B) = 1$. Thus we can* separate points by continuous functions[1].

Proof. The function $x \mapsto \frac{\rho(x,A)}{\rho(x,A)+\rho(x,B)}$ fits the bill. □

Exercise 1.15. *Let $F \subset \mathcal{M}$ be closed. The (uniformly) continuous function $f_\epsilon(x) = \left(1 - \frac{\rho(x,F)}{\epsilon}\right)_+$ satisfies*

$$f_\epsilon(x) = \begin{cases} 1 & \text{if } x \in F, \\ 0 & \text{if } x \notin F_\epsilon, \end{cases}$$

where $F_\epsilon = \{y \in M \mid \rho(y, F) < \epsilon\}$ is the ϵ-enlargement of F. Note also that $I_F(x) \leq f_\epsilon(x) \leq I_{F_\epsilon}(x)$. Construct another such function using the proof of Urysohn's lemma.

1.4 Limit points and accumulation points

Let $X \subset \mathcal{M}$.

- $x \in X$ is called a *limit point* of X if, for any open neighbourhood U_x of x, $U_x \cap X \neq \emptyset$.
- $x \in X$ is called an *accumulation point* (also called cluster point) of X if, for any open neighbourhood U_x of x, $(U_x \setminus \{x\}) \cap X \neq \emptyset$.
- Limit points, which are not accumulation points are called *isolated points*.
- A set is closed if and only if it contains all its limit points (**exercise**).
- A set is called *discrete* if all its points are isolated points.
- A closed set is called *perfect* if all its points are accumulation points.

Exercise 1.16. *Give examples of each of these in $\mathbb{R}^d$ (equipped with the standard metric).*

Exercise 1.17. *Every subset of a discrete metric space is discrete. Every discrete set is closed.*

Exercise 1.18. *Can a discrete subset of $\mathbb{R}^d$ set be uncountable? Can a perfect subset of $\mathbb{R}^d$ be countable?*

[1] Topological spaces where one can do this are called *completely Hausdorff (or completely T_2)* spaces.

1.5 Denseness

$X \subset \mathcal{M}$ is *dense* in $\mathcal{M}$ if, for all $x \in \mathcal{M}$, and for any open neighbourhood U_x of x, $U_x \cap X \neq \emptyset$. A metric space is called *separable* if it contains a countable dense subset.

Example 1.19. *$\mathbb{R}^d$ is separable because it contains the countable dense subset $\mathbb{Q}^d$.*

Exercise 1.20. *Show that $C[0,1]$ with the uniform metric is separable. (Hint: Stone-Weierstrass!)*

Why do we care about separability? Well, measurability plays well with respect to countable operations. So if one wants to do measure theory on metric spaces, separability helps to restrict attention to countable operations (such as taking sup over a countable dense set).

1.6 Convergence, Cauchy sequences and completeness

Note that every metric space $\mathcal{M}$ is *Hausdorff* (also called T_2), i.e. any two distinct points can be separated by disjoint open sets, i.e., $\exists$ neighbourhoods U_x, U_y of $x, y \in \mathcal{M}$ such that $U_x \cap U_y = \emptyset$. One can thus define the concept of convergence of a sequence.

- Let $\{x_n\}$ be a sequence in $\mathcal{M}$. We say that *x_n converges to x (written $x_n \to x$)*, if, for any neighbourhood U_x of x, $\exists N \geq 1$, such that for all $n \geq N$, $x_n \in U_x$). Note that how this encompasses the (ϵ, N)-definition of convergence. The Hausdorff property ensures that the limit of a sequence is unique.
- Let $\{x_n\}$ be a sequence in $\mathcal{M}$. We say that x_n is *Cauchy* if, for any $\epsilon > 0$, $\exists N \geq 1$ such that for all $m, n \geq N$, $\rho(x_m, x_n) < \epsilon$.

Exercise 1.21. *Show that all convergent sequences are Cauchy. Give a counterexample to the converse statement.*

Exercise 1.22. *What are all the Cauchy sequences in a discrete metric space?*

$X \subset \mathcal{M}$ is called *complete* if all Cauchy sequences in X converge to some element of x. For example, $\mathbb{R}^d$ is complete.

Exercise 1.23. *Any discrete metric space is complete.*

Exercise 1.24. *$C[0,1]$ is complete in the uniform metric.*

Exercise 1.25. *Any closed subset of a complete metric space is itself complete.*

The fact that $C[0,1]$ is complete with the uniform metric will be very useful to us. Complete, separable metric spaces are also called *Polish* spaces. Thus $C[0,1]$ is a Polish space under the uniform metric.

Exercise 1.26. *$f : \mathcal{M}_1 \to \mathcal{M}_2$ is continuous if and only if for any sequence $\{x_n\}$ in $\mathcal{M}_1$ converging to x, the sequence $\{f(x_n)\}$ in $\mathcal{M}_2$ converges to $f(x)$.*

1.7 Compactness

$X \subset \mathcal{M}$ is called *compact* if any open cover $\{U_\alpha\}_{\alpha \in I}$ admits a finite subcover. X is called *bounded* if $\exists$ an open ball $B(x, \epsilon) \supset X$.

Exercise 1.27. *Let* $\operatorname{diam}(A) := \sup_{x,y \in A} \rho(x, y)$ *be the* diameter *of* A. *Show that* A *is bounded if and only if* A *is either empty or* $\operatorname{diam}(A)$ *is finite.*

Exercise 1.28. *If* X *is compact, then it is closed and bounded. In Euclidean spaces the converse also holds (the so-called Heine-Borel theorem).*

Exercise 1.29. *What are the compact subsets of a discrete metric space?*

Exercise 1.30. *Suppose* $K \subset \mathcal{M}_1$ *is compact and* $f : \mathcal{M}_1 \to \mathcal{M}_2$ *is continuous, then* f *is uniformly continuous.*

Exercise 1.31. *Suppose* $K \subset \mathcal{M}_1$ *is compact and* $f : \mathcal{M}_1 \to \mathcal{M}_2$ *is continuous, then* $f(K)$ *is compact (hence bounded).*

$X \subset \mathcal{M}$ is called *totally bounded*, if, for any $\epsilon > 0$, $\exists$ points $x_1, \ldots, x_k \in \mathcal{M}$, where $k = k(\epsilon) \geq 1$, such that $X \subset \cup_{i=1}^{k} B(x_k; \epsilon)$[2].

Exercise 1.32. *Prove that the closure of a totally bounded set it totally bounded.*

For metric spaces there are equivalent characterisations of compactness.

Lemma 1.33. *The following are equivalent:*

(a) *K is compact.*

(b) *K is complete and totally bounded.*

(c) *Every infinite subset of K has an accumulation point.*

(d) *Every sequence in K has a convergent subsequence (this is also called* sequential compactness*).*

A set $X \subset \mathcal{M}$ is called *precompact* or *relatively compact* if $\bar{X}$ is compact.

Exercise 1.34. *In complete metric spaces, the precompact sets are precisely the totally bounded sets.*

Exercise 1.35. *Is the closed ball* $B[x; \epsilon]$ *compact?*

As we will be working with $C[0, 1]$ extensively, let us characterize its compact subsets. For this we need to define the notion of *equicontinuity* of a family of functions.

Let $\mathcal{F}$ be a family of functions from $\mathcal{M}_1$ to $\mathcal{M}_2$. $\mathcal{F}$ is called *equicontinuous at $x \in \mathcal{M}_1$*, if, for any $\epsilon > 0$, $\exists \delta = \delta(x, \epsilon)$ such that $\rho_1(y, x) < \delta \implies \rho_2(f(y), f(x)) < \epsilon$ for all $f \in \mathcal{F}$. If $\mathcal{F}$ is equicontinuous at all $x \in \mathcal{M}_1$, then we simply say that the family $\mathcal{F}$ is equicontinuous.

Exercise 1.36. *Define the notion of* uniform equicontinuity *and then show that if* $\mathcal{M}_1$ *is compact, then* $\mathcal{F}$ *is uniformly equicontinuous.*

[2] Such a finite collection of ϵ-balls covering a space is called an *ϵ-net*. We will encounter them time and again when we start talking about empirical process theory.

Exercise 1.37. *Let M_1 be compact and $f : \mathcal{M}_1 \times \mathcal{M}_1 \to \mathcal{M}_2$ be continuous. Then the family $\mathcal{F} = \{f_y(x) = f(x,y) \mid y \in \mathcal{M}_1\}$ is uniformly equicontinuous.*

Exercise 1.38. *Let $M > 0$. Let $\mathcal{F}$ be a family of functions from a convex and open set $X \subset \mathbb{R}^d$ to $\mathbb{R}^{d'}$ such that $\|Df(x)\| \leq M$ for all $x \in X$ and for all $f \in \mathcal{F}$. Then $\mathcal{F}$ is uniformly equicontinuous.*

Theorem 1.39 (Arzelà-Ascoli). *Let $K \subset \mathcal{M}$ be compact. Consider the metric space $C(K)$ of all continuous functions on K with the uniform metric. $S \subset C(K)$ is compact if and only if S is closed, bounded and equicontinuous.*

The *modulus of continuity* at level $\delta > 0$ of a function $x : \mathcal{M}_1 \to \mathcal{M}_2$ is defined as

$$w_x(\delta) = \sup_{\rho_1(s,t) \leq \delta} \rho_2(x(s), x(t)).$$

Theorem 1.40 (Arzelà-Ascoli for $C[0,1]$). *$S \subset C([0,1])$ has compact closure if and only if the following two conditions hold:*

1. $\sup_{x \in S} |x(0)| < \infty$.
2. $\lim_{\delta \to 0} \sup_{x \in S} w_x(\delta) = 0$.

1.8 Connectedness

Do not need now.

2 Probability measures on metric Spaces

Let $(\mathcal{M}, \rho)$ be a metric space. Consider the Borel σ-field $\mathcal{B}$ of $(\mathcal{M}, \rho)$ (i.e. the σ-field generated by all open sets). We will be interested in defining a notion of convergence in distribution on $(\mathcal{M}, \rho)$. Our main reference for this section would be Billingsley (2013) and also the previous editions of that book.

2.1 Preliminaries

Lemma 2.1 (Regularity of probability measures). *Every probability measure μ on $(\mathcal{M}, \mathcal{B})$ is regular, i.e. for any Borel set A, and for any $\epsilon > 0$, one can find G open and F closed such that $F \subset A \subset G$, and $\mu(G \setminus F) < \epsilon$.*

Proof. Use the "good set" principle. Show that statement is true if A is a closed set, and then show that the set of all such sets forms a σ-field. Fill in the details. □

Notation: Let μ be a measure and $f \in C_b(\mathcal{M})$. Then $\mu f := \int f \, d\mu$. For probability measures $\mathbb{P}$, $\mathbb{P}f$ is thus same as $\mathbb{E}(f(X))$ where $X \sim \mathbb{P}$.

A class of functions $\mathcal{F}$ on $(\mathcal{M}, \mathcal{B})$ is called *(probability) measure determining* if $\mu f = \nu f, \forall f \in \mathcal{F}$ implies $\mu = \nu$.

Lemma 2.2. *$\{I_F \mid F$ closed$\}$ is a probabability determining class. Same goes for $\{I_G \mid G$ open$\}$.*

Proof. For a general Borel set A, by regularity, one can construct two sequences of closed sets $F_{in} \subset A$ such that $\mathbb{P}_i(F_{in}) \to \mathbb{P}_i(A), i = 1,2$. But then $\mathbb{P}_i(F_{1n} \cup F_{2n}) \to \mathbb{P}_i(A), i = 1,2$, which implies that $\mathbb{P}_1(A) = \mathbb{P}_2(A)$. □

Lemma 2.3. *$C_b(\mathcal{M})$, the class of bounded continuous functions on $\mathcal{M}$ is probability determining. Same goes for $C_{bu}(\mathcal{M})$, the class of bounded uniformly continuous functions on $\mathcal{M}$.*

Proof. For every closed F, approximate I_F by f_ϵ to get $\mathbb{P}_i(F) \leq \mathbb{P}_i f_\epsilon \leq \mathbb{P}_i(F_\epsilon), i = 1,2$. But $\mathbb{P}_i(F_\epsilon) \downarrow \mathbb{P}_i(F)$ with $\epsilon \downarrow 0, i = 1,2$, and $\mathbb{P}_1 f_\epsilon = \mathbb{P}_2 f_\epsilon$. This shows that $\mathbb{P}_1(F) = \mathbb{P}_2(F)$. As the functions I_F are probability determining, we are done. □

Lemma 2.4. *Any π-system $\mathcal{A}$ generating the Borel σ-field $\mathcal{B}$ of $\mathcal{M}$ is a determining class of sets.*

Proof. The class $\mathcal{S} = \{A \in \mathcal{B} \mid \mathbb{P}(A) = \mathbb{Q}(A)\}$ is a λ-system containing the π-system $\mathcal{A}$. Hence the desired conslusion follows from the π-λ theorem. □

A probability measure $\mathbb{P}$ on $(\mathcal{M}, \mathcal{B})$ is called *tight*, if, for any $\epsilon > 0$, there exists a compact set $K \subset \mathcal{M}$ such that $\mathbb{P}(K) > 1 - \epsilon$.

Exercise 2.5. *Probability measures on σ-compact metric spaces (i.e. metric spaces that are countable union of compact sets) are tight.*

Lemma 2.6. *Probability measures on Polish spaces are tight.*

Proof. Let $\{x_1, x_2, \ldots\}$ be countable dense subset of $\mathcal{M}$. Let $U_{ik} = B(x_i; \frac{1}{k})$. Then $\cup_{i \geq 1} U_{ik}$ covers M for any $k \geq 1$. Choose $n_k \geq 1$ such that $\mathbb{P}(\cup_{1 \leq i \leq n_k} U_{ik}) > 1 - \frac{\epsilon}{2^k}$. Now set $A = \cap_{k \geq 1}(\cup_{1 \leq i \leq n_k} U_{ik})$. Note that $\mathbb{P}(A^c) = \mathbb{P}(\cup_{k \geq 1}(\cup_{1 \leq i \leq n_k} U_{ik})^c) \leq \sum_{k \geq 1} \mathbb{P}((\cup_{1 \leq i \leq n_k} U_{ik})^c) \leq \sum_{k \geq 1} \frac{\epsilon}{2^k} = \epsilon$. Hence $\mathbb{P}(A) > 1 - \epsilon$.

Note also that, by its very construction, A is totally bounded. As $\mathcal{M}$ is complete, $\bar{A}$ is complete. But the closure of a totally bounded set is also totally bounded. Hence, A is compact. But, $\mathbb{P}(\bar{A}) \geq \mathbb{P}(A) > 1 - \epsilon$, which completes the proof. □

2.2 Weak convergence

By $\mathcal{P}(\mathcal{M})$ we will denote the set of all probability measures on $\mathcal{M}$. Let $\mathbb{P}, \mathbb{P}_n, n \geq 1, \in \mathcal{P}(M)$. We say that *$\mathbb{P}_n$ converges weakly to $\mathbb{P}$, denoted as $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$*, if $\mathbb{P}_n f \to \mathbb{P} f, \forall f \in C_b(\mathcal{M})$. The following lemma gives important equivalent characterizations.

Lemma 2.7 (Portmanteau). *The following are equivalent:*

(a) $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$.

(b) $\mathbb{P}_n f \to \mathbb{P} f, \forall f \in C_{ub}(\mathcal{M})$.

(c) $\limsup_{n \to \infty} \mathbb{P}_n(F) \leq \mathbb{P}(F), \forall F$ *closed.*

(d) $\liminf_{n \to \infty} \mathbb{P}_n(G) \geq \mathbb{P}(G), \forall G$ *open.*

(e) $\mathbb{P}_n(A) \to \mathbb{P}(A), \forall A$ *such that* $\mathbb{P}(\partial A) = 0$[3].

Proof. "(a) $\implies$ (b)": Obvious!

"(b) $\implies$ (c)": We have, for any $\epsilon > 0$,

$$\limsup \mathbb{P}_n(F) \leq \limsup \mathbb{P}_n f_\epsilon = \lim \mathbb{P}_n f_\epsilon = \mathbb{P} f_\epsilon \leq \mathbb{P}(F_\epsilon).$$

Now let $\epsilon \downarrow 0$.

"(c) $\iff$ (d)": For the " $\implies$ " direction, note that, by (b) $\limsup \mathbb{P}_n(G^c) \leq \mathbb{P}(G^c)$, i.e. $\limsup(1 - \mathbb{P}_n(G)) \leq 1 - \mathbb{P}(G)$, i.e. $1 - \liminf \mathbb{P}_n(G) \leq 1 - \mathbb{P}(G)$, which is same as what we want. Similarly, "(d) $\implies$ (c)".

"(c) $\implies$ (e)": By (c) and (d),

$$\mathbb{P}(\bar{A}) \geq \limsup \mathbb{P}_n(\bar{A}) \geq \limsup \mathbb{P}_n(A) \geq \liminf \mathbb{P}_n(A) \geq \liminf \mathbb{P}_n(A^\circ) \geq \mathbb{P}(A^\circ).$$

Now $\mathbb{P}(\partial A) = 0$ implies that $\mathbb{P}(\bar{A}) = \mathbb{P}(A^\circ) = \mathbb{P}(A)$ and hence we are done.

"(e) $\implies$ (a)": Let $f \in C_b(\mathcal{M})$. By scaling and translating suitably, we may assume that $0 < f < 1$. Now $\mathbb{P} f = \int_0^1 \mathbb{P}(f > t)\, dt$. Note that $\mathbb{P}(f > t) = 1 - F_{f(X)}(t)$, where $F_{f(X)}$ is the CDF of the (real-valued) random variable $f(X)$. As the set of jumps of a CDF is at most countable, and $\partial\{f > t\} \subset \{f = t\}$ (by continuity of f), we conclude that $\mathbb{P}_n(f > t) \to \mathbb{P}(f > t)$ for all but countably many points. Hence an application of BCT gives that $\mathbb{P}_n f = \int_0^1 \mathbb{P}_n(f > t)\, dt \to \int_0^1 \mathbb{P}(f > t)\, dt = \mathbb{P} f$. □

Recall the statement $x_n \to x$ if and only if every subsequence of x_n contains a further subsequence that converges to x. We have an analogue of this for weak convergence.

Lemma 2.8. $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$ *if and only if every subsequence of* $\mathbb{P}_n$ *contains a further subsequence that converges weakly to* $\mathbb{P}$.

Proof. Only the "if" part is non-trivial. Let $f \in C_b(\mathcal{M})$. Consider the real-sequence $\{\mathbb{P}_n f\}$. Take a subsequence $\{\mathbb{P}_{n_k} f\}$. By hypothesis, for the subsequence $\{\mathbb{P}_{n_k}\}$ of $\{\mathbb{P}_n\}$, there exists a further subsequence $\{\mathbb{P}_{n_{k_\ell}}\}$ that converges weakly to $\mathbb{P}$. Hence $\{\mathbb{P}_{n_{k_\ell}} f\}$ converges to $\mathbb{P} f$. So any subsequence $\{\mathbb{P}_{n_k} f\}$ of $\{\mathbb{P}_n f\}$ contains a further subsequence $\{\mathbb{P}_{n_{k_\ell}} f\}$ that converges to $\mathbb{P} f$. It follows that $\mathbb{P}_n f \to \mathbb{P} f$, and we are done. □

Exercise 2.9. *Show that* $\delta_{x_n} \xrightarrow{w} \delta_x$ *if and only if* $x_n \to x$.

Exercise 2.10. *Show that, if* $\delta_{x_n} \xrightarrow{w} \mathbb{P}$, *then* $\mathbb{P} = \delta_x$ *for some* x.

A class of sets $\mathcal{A} \subset \mathcal{B}$ is called *convergence determining* if $\mathbb{P}_n(A) \to \mathbb{P}(A)$ for all $\mathbb{P}$-continuity sets $A \in \mathcal{A}$ implies that $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$.

Exercise 2.11. *Any convergence determining class is a probability determining class.*

Lemma 2.12. *Let* $\mathcal{A}$ *be a* π*-system such that every open set* G *can be written as a countable union of sets from* $\mathcal{A}$. *Then* $\mathbb{P}_n(A) \to \mathbb{P}(A)$ *for all* $A \in \mathcal{A}$ *implies that* $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$.

[3] Such sets A are called $\mathbb{P}$-continuity sets.

Lemma 2.13. *Suppose $\mathcal{M}$ is separable. Let $\mathcal{A}$ be a π-system such that for any $x \in \mathcal{M}$ and $\epsilon > 0$, there exists $A_{x,\epsilon} \in \mathcal{A}$ with $x \in A^\circ_{x,\epsilon} \subset A_{x,\epsilon} \subset B(x;\epsilon)$. Then $\mathbb{P}_n(A) \to \mathbb{P}(A)$ for all $A \in \mathcal{A}$ implies that $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$.*

Let $\mathcal{A}_{x,\epsilon}$ be the class of sets A such that $x \in A^\circ \subset A \subset B(x;\epsilon)$. Let $\partial\mathcal{A}_{x,\epsilon} = \{\partial A \mid A \in \mathcal{A}_{x,\epsilon}\}$.

Lemma 2.14. *Suppose $\mathcal{M}$ is separable. Let $\mathcal{A}$ be a π-system such that $\partial\mathcal{A}_{x,\epsilon}$ either contains $\emptyset$ or has uncountably many disjoint elements. Then $\mathcal{A}$ is a convergence determining class.*

Example 2.15. *On $\mathbb{R}^d$, rectangles of the form $(a,b] = (a_1,b_1] \times \cdots \times (a_d,b_d]$ form a convergence determining class.*

Example 2.16. *On $\mathbb{R}^\infty$, finite dimensional cylinders $\pi^{-1}_{i_1,\ldots,i_d}((a,b])$ form a convergence determining class.*

Example 2.17. *On $C[0,1]$, finite dimensional cylinders $\pi^{-1}_{t_1,\ldots,t_k}((a,b])$* do not *form a convergence determining class. However, they do form a determining class. To see this note that the sequence of functions $z_n(t) = nt\mathbb{I}(t \le n) + (2-nt)\mathbb{I}(n < t \le 2n)$ converge pointwise to the 0 function. However, $\|z_n\|_\infty = 1$. So δ_{z_n} does not converge weakly to δ_0. However, since $\pi^{-1}_{t_1,\ldots,t_k} z_n = (z_n(t_1),\ldots,z_n(t_k)) = (0,\ldots,0) = \pi^{-1}_{t_1,\ldots,t_k}\delta_0$, finite dimensional distributions of δ_{z_n} converge δ_0 (prove this rigorously).*

2.3 The mapping theorem and Skorohod's representation

As is true for random variables, if h is a continuous mapping $\mathcal{M}_1 \to \mathcal{M}_2$, and $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$, then $\mathbb{P}_n \circ h^{-1} \xrightarrow{w} \mathbb{P} \circ h^{-1}$. In fact, we can say something stronger.

Theorem 2.18 (The mapping theorem). *Let $h : \mathcal{M}_1 \to \mathcal{M}_2$ be such that $\mathbb{P}(D_h) = 0$, where D_h is the set of discontinuity points of h. Then $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$ implies $\mathbb{P}_n \circ h^{-1} \xrightarrow{w} \mathbb{P} \circ h^{-1}$.*

Proof. Let F be a closed subset of $\mathcal{M}_2$. Note that

$$\begin{aligned}
\limsup_{n\to\infty} \mathbb{P}_n \circ h^{-1}(F) &\le \limsup_{n\to\infty} \mathbb{P}_n(\overline{h^{-1}(F)}) \\
&\le \mathbb{P}(\overline{h^{-1}(F)}) \\
&= \mathbb{P}(\overline{h^{-1}(F)} \cap D_h^c) \\
&\le \mathbb{P} \circ h^{-1}(F),
\end{aligned}$$

where the last inequality follows from the set inequality $\overline{h^{-1}(F)} \cap D_h^c \subset h^{-1}(F)$, which is true beacuse, if x belongs to the LHS, then there is a sequence x_n such that $h(x_n) \in F$ and $x_n \to x$. But, since $x \in D_h^c$, this means that $h(x_n) \to h(x)$. As F is closed, $h(x) \in F$, i.e. $x \in h^{-1}(F)$. □

Exercise 2.19. *Another tool for your portmanteau: $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$ if and only if for any bounded measurable function h with $\mathbb{P}(D_h) = 0$ one has $\mathbb{P}_n h \to \mathbb{P}h$.*

Exercise 2.20. *Using the previous exercise, give an alternative proof of the mapping theorem.*

There is a theorem of Skorohod which constructs random elements X_n, X on a common probability space such that $X_n \sim \mathbb{P}_n, X \sim \mathbb{P}$ and $X_n \to X$ almost surely. For this $\mathbb{P}$ needs to have separable support. This result is very useful in practice beacuse almost sure convergence is easy to work with and implies weak convergence.

Before we embark on a proof in the gerenal case, let us prove it for random variables.

Lemma 2.21. *Let F_n, F be univariate CDFs with $F_n \xrightarrow{d} F$. Then there exist random variables $X_n \sim F_n, X \sim F$ defined on the same probability space $(\Omega, \mathcal{A}, Q)$ such that $X_n \xrightarrow{a.s.} X$.*

2.4 Relative compactness and Prohorov's theorem

A family Π of probaiblity measures is called *relatively compact* if any sequence $\{\mathbb{P}_n\}$ in Π has a weakly convergent subsequence. Thus, a sequence $\{\mathbb{P}_n\}$ is relatively compact if any subsequence $\{\mathbb{P}_{n_k}\}$ has a further subsequence $\{\mathbb{P}_{n_{k_\ell}}\}$ that converges weakly.

Lemma 2.22. *Consider probability measures on $C[0,1]$. If $\{\mathbb{P}_n\}$ is relatively compact, and all finite dimensional distributions converge weakly, then there exists a probability measure $\mathbb{P}$ on $C[0,1]$ such that $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$.*

Proof. Consider a subsequence $\{\mathbb{P}_{n_\ell}\}$ converging to some probability measure $\mathbb{Q}$. Note then that $\mathbb{P}_{n_\ell} \circ \pi^{-1}_{t_1,\ldots,t_k} \xrightarrow{w} \mathbb{Q} \circ \pi^{-1}_{t_1,\ldots,t_k}$. It follows that $\mathbb{P}_n \circ \pi^{-1}_{t_1,\ldots,t_k} \xrightarrow{w} \mathbb{Q} \circ \pi^{-1}_{t_1,\ldots,t_k}$. As finite dimensional cylinders form a determining class, it also follows that all subsequential weak limits of $\mathbb{P}_n$ are equal to $\mathbb{Q}$. Since, by relative compactness, any subsequence has a further subsequence that converges weakly to $\mathbb{Q}$, we conclude that $\mathbb{P}_n \xrightarrow{w} \mathbb{Q}$. □

How does one show relative compactness of a family? A sufficient condition is *tightness* of the family—a family Π is called tight if for any $\epsilon > 0$ there exists a compact set K_ϵ such that $\mathbb{P}(K_\epsilon) > 1 - \epsilon$ for all $\mathbb{P} \in \Pi$.

Theorem 2.23 (Prohorov). *If a family Π is tight, then it is relatively compact.*

Proof. To be written up. □

Theorem 2.24. *Suppose that $\mathcal{M}$ is Polish. Then relative compactness of a family Π implies its tightness.*

Proof. The proof is similar to the proof of Lemma 2.6, which establishes tightness of a single measure. It suffices to show that if $\mathcal{M}$ can be written as an increasing union of open sets G_i, then for any $\epsilon > 0$ there exists $n_\epsilon > 0$ such that $\mathbb{P}(G_{n_\epsilon}) > 1 - \epsilon$, for all $\mathbb{P} \in \Pi$. Assume not, then, for some $\epsilon_0 > 0$, we can find a sequence $\{\mathbb{P}_n\}$ in Π such that $\mathbb{P}_n(G_n) \leq 1 - \epsilon_0$. By relative compactness, there is a further subsequence $\{\mathbb{P}_{n_k}\}$ converging weakly to some probability measure $\mathbb{Q}$. Now for any $m \geq 1$,

$$\mathbb{Q}(G_m) \leq \liminf \mathbb{P}_{n_k}(G_m) \leq \liminf \mathbb{P}_{n_k}(G_{n_k}) \leq 1 - \epsilon_0.$$

Let $m \to \infty$ to get $\mathbb{Q}(\mathcal{M}) \leq 1 - \epsilon_0$, a contradiction.

To use this, write $M = \cup_{n \geq 1} B(x_n; 1/k)$, where $\{x_1, x_2, \ldots\}$ is a countable dense set. There is n_k such that $\mathbb{P}(\cup_{n=1}^{n_k} B(x_n; 1/k)) > 1 - \frac{\epsilon}{2^k}$ for all $\mathbb{P} \in \Pi$. Hence, the totally bounded set $A = \cap_{k \geq 1} \cup_{n=1}^{n_k} B(x_n; 1/k)$ satisfies $\mathbb{P}(A) > 1 - \epsilon$ for all $\mathbb{P} \in \Pi$. $\bar{A}$ is the required compact set. □

3 Weak convergence in $C[0,1]$

3.1 Tightness in $C[0,1]$

By Arzelà-Ascoli we can characterize compact sets of $C[0,1]$. This enables us to formulate an equivalent criterion for tightness.

Lemma 3.1. *Let $\mathbb{P}_n$ be a sequence of Borel probability measures on $C[0,1]$. Then $\{\mathbb{P}_n\}$ is tight if and only if the following two conditions hold:*

(i) For all $\eta > 0$, $\exists a \geq 0$ such that, for all $n \geq 1$, one has

$$\mathbb{P}_n(|x(0)| \geq a) \leq \eta.$$

Note that this condition is equivalent to saying that $\{\mathbb{P}_n \circ \pi_0^{-1}\}$ is tight.

(ii) For all $\eta > 0, \epsilon > 0$, $\exists \delta \in (0,1), n_0 \geq 1$ such that, for all $n \geq n_0$, one has

$$\mathbb{P}_n(w_x(\delta) \geq \epsilon) \leq \eta.$$

Now we already know that tightness plus convergence of finite dimensional distributions implies weak convergence. Combining this with the above lemma we can say the following.

Lemma 3.2. *Let $\mathbb{P}_n$ be a sequence of Borel probability measures on $C[0,1]$. Suppose*

(i) $\mathbb{P}_n \circ \pi_{t_1,\ldots,t_k}^{-1}$ convereges weakly, for any $t_1, \ldots, t_k \in [0,1], k \geq 1$.

(ii) For all $\epsilon > 0$

$$\lim_{\delta \to 0} \limsup_{n \to \infty} \mathbb{P}_n(w_x(\delta) \geq \epsilon) = 0.$$

Then there is a Borel probability measure $\mathbb{P}$ on $C[0,1]$ such that $\mathbb{P}_n \xrightarrow{w} \mathbb{P}$. The converse is also true.

3.2 Wiener measure and Donsker's theorem

Let ξ_i be i.i.d. with $\mathbb{E}\xi = 0, \mathbb{E}\xi^2 = \sigma^2$, living in some probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $S_n = \sum_{i=1}^n \xi_i, S_0 \equiv 0$. Construct a random function $X_n : (\Omega, \mathcal{A}, \mathbb{P}) \to (C[0,1], \mathcal{B}(C[0,1]))$ as follows

$$X_n(\omega)(t) \equiv X_n(\omega, t) = \begin{cases} \frac{S_i(\omega)}{\sigma\sqrt{n}}, & \text{if } t = \frac{i}{n}, \\ \alpha \frac{S_i(\omega)}{\sigma\sqrt{n}} + (1-\alpha)\frac{S_{i+1}(\omega)}{\sigma\sqrt{n}}, & \text{if } t = \alpha\frac{i}{n} + (1-\alpha)\frac{i+1}{n}, 0 \leq i < n. \end{cases}$$

If not necessary, we will drop ω for notational simplicity. Note that we can compactly write $X_n(t)$ as

$$X_n(t) = \frac{S_{\lfloor nt \rfloor}}{\sigma\sqrt{n}} + (nt - \lfloor nt \rfloor)\frac{\xi_{\lfloor nt \rfloor + 1}}{\sigma\sqrt{n}}.$$

Exercise 3.3. *Show that X_n defined above is indeed a random function, i.e., X_n is $\mathcal{A}/\mathcal{B}(C[0,1])$ measurable.*

We will show that $\{\mathbb{P}_n = \mathbb{P} \circ X_n^{-1}\}$ is tight and $\mathbb{P}_n \circ \pi_{t_1,\ldots,t_k}^{-1}$ convereges weakly to a certain k-variate Gaussian, for any $t_1, \ldots, t_k \in [0,1], k \geq 1$. This will establish the existence of a measure on $C[0,1]$ with Gaussian finite dimensional distributions.

Theorem 3.4 (Existence of Wiener measure and Donsker's theorem). *There exists, on $C[0,1]$, a Borel probability measure W such that, for any $t_1, \ldots, t_k \in [0,1], k \geq 1$, one has $W \circ \pi^{-1}_{t_1,\ldots,t_k} \equiv \mathcal{N}(0, \Sigma^{(t_1,\ldots,t_k)})$, where $\Sigma^{(t_1,\ldots,t_k)}_{ij} = t_i \wedge t_j$. Moreover, $\mathbb{P}_n \xrightarrow{w} W$*[a].

[a] W is called the Wiener measure after mathematician Norbert Wiener, who was the first to construct Brownian motion rigorously.

Proof. Note that $X_n(0) = 0$. For $t_1, t_2, \ldots, t_k \in (0,1], k \geq 1$,

$$\begin{aligned}
&(X_n(t_1), X_n(t_2) - X_n(t_1)), \ldots, X_n(t_k) - X_n(t_{k-1})) \\
&\quad = \left(\frac{S_{\lfloor nt_1 \rfloor}}{\sigma\sqrt{n}} + o_{\mathbb{P}}(1), \frac{S_{\lfloor nt_2 \rfloor} - S_{\lfloor nt_1 \rfloor}}{\sigma\sqrt{n}} + o_{\mathbb{P}}(1), \ldots, \frac{S_{\lfloor nt_k \rfloor} - S_{\lfloor nt_{k-1} \rfloor}}{\sigma\sqrt{n}} + o_{\mathbb{P}}(1) \right) \\
&\quad \xrightarrow[CLT]{d} (Y_1, \ldots, Y_k),
\end{aligned}$$

where the $Y_j \sim \mathcal{N}(0, t_j - t_{j-1}), t_0 \equiv 0$, are independent. If we interpret $X(0) = 0 \equiv \mathcal{N}(0,0)$, then the above is true for any $t_1, t_2, \ldots, t_k \in [0,1], k \geq 1$. It follows that (prove)

$$\mathbb{P}_n \circ \pi^{-1}_{t_1,\ldots,t_k} \xrightarrow{w} \mathcal{N}(0, \Sigma^{(t_1,\ldots,t_k)}),$$

where $\Sigma^{(t_1,\ldots,t_k)}_{ij} = t_i \wedge t_j$. This verifies condition (i) of Lemma 3.2.

Let $0 = t_0 < t_1 < \ldots t_{k-1} < t_k = 1$ be *δ-sparse*, i.e. $\min_{1<i<k} |t_i - t_{i-1}| \geq \delta$. Then

$$w_x(\delta) \leq 3 \max_{1 \leq i \leq k} \sup_{t_{i-1} \leq s \leq t_i} |x(s) - x(t_{i-1})|.$$

It follows that

$$\mathbb{P}_n(w_x(\delta) \geq \epsilon) \leq \sum_{1 \leq i \leq k} \mathbb{P}_n(\sup_{t_{i-1} \leq s \leq t_i} |x(s) - x(t_{i-1})| \geq \frac{\epsilon}{3}).$$

For simplicity, we choose a particular equispaced δ-sparse set: $t_i = \frac{im}{n}, 0 \leq i < k, m \in \mathbb{N}, t_k = 1$. Then we must have $\delta \leq t_2 - t_1 = \frac{m}{n}$. So, in particular, we can take $m = \lceil n\delta \rceil$. Also, $1 > t_{k-1} = (k-1)\frac{m}{n}$. If we choose the maximum possible number of points t_i, then $1 \leq \frac{km}{n}$. Then $k = \lceil \frac{n}{m} \rceil \to \frac{1}{\delta}$.

Now note that

$$\begin{aligned}
\sup_{t_{i-1} \leq s \leq t_i} |X_n(s) - X_n(t_{i-1})| &= \max_{nt_{i-1} \leq j < nt_i} \sup_{0 \leq \alpha \leq 1} \left| \alpha \frac{S_j}{\sigma\sqrt{n}} + (1-\alpha)\frac{S_{j+1}}{\sigma\sqrt{n}} - \frac{S_{nt_{i-1}}}{\sigma\sqrt{n}} \right| \\
&= \max_{nt_{i-1} \leq j < nt_i} \max \left\{ \frac{|S_j - S_{nt_{i-1}}|}{\sigma\sqrt{n}}, \frac{|S_{j+1} - S_{nt_{i-1}}|}{\sigma\sqrt{n}} \right\} \\
&= \max_{nt_{i-1} \leq j \leq nt_i} \frac{|S_j - S_{nt_{i-1}}|}{\sigma\sqrt{n}}.
\end{aligned}$$

By stationarity of the sequence ξ_i,

$$\max_{nt_{i-1} \leq j \leq nt_i} \frac{|S_j - S_{nt_{i-1}}|}{\sigma\sqrt{n}} \stackrel{d}{=} \max_{1 \leq j \leq m} \frac{|S_j|}{\sigma\sqrt{n}},$$

for any $1 \le i < k$. For $i = k$,

$$\max_{nt_{i-1} \le j \le nt_i} \frac{|S_j - S_{nt_{i-1}}|}{\sigma\sqrt{n}} \stackrel{d}{=} \max_{1 \le j \le m'} \frac{|S_j|}{\sigma\sqrt{n}},$$

where $m' = n - (k-1)m \le m$. All in all, we conclude that

$$\begin{aligned}
\mathbb{P}_n(w_x(\delta) \ge \epsilon) &\le \sum_{1 \le i \le k} \mathbb{P}_n(\sup_{t_{i-1} \le s \le t_i} |x(s) - x(t_{i-1})| \ge \frac{\epsilon}{3}) \\
&= \sum_{1 \le i \le k} \mathbb{P}(\sup_{t_{i-1} \le s \le t_i} |X_n(s) - X_n(t_{i-1})| \ge \frac{\epsilon}{3}) \\
&= (k-1)\mathbb{P}\left(\max_{1 \le j \le m} \frac{|S_j|}{\sigma\sqrt{n}} \ge \frac{\epsilon}{3}\right) + \mathbb{P}\left(\max_{1 \le j \le m'} \frac{|S_j|}{\sigma\sqrt{n}} \ge \frac{\epsilon}{3}\right) \\
&\le k\mathbb{P}\left(\max_{1 \le j \le m} \frac{|S_j|}{\sigma\sqrt{n}} \ge \frac{\epsilon}{3}\right) \\
&= k\mathbb{P}\left(\max_{1 \le j \le m} |S_j| \ge \frac{\epsilon\sigma\sqrt{n}}{3}\right) \\
&\le 3k \max_{1 \le j \le m} \mathbb{P}\left(|S_j| \ge \frac{\epsilon\sigma\sqrt{n}}{9}\right) \quad \text{(by Etemadi's maximal inequality[4])} \\
&\le \frac{6}{\delta} \max_{1 \le j \le m} \mathbb{P}\left(|S_j| \ge \frac{\epsilon\sigma\sqrt{m}}{9\sqrt{2\delta}}\right),
\end{aligned}$$

where the last inequality holds for all large n, since $\frac{1}{2\delta} < \frac{1}{\delta} = \lim_{n\to\infty} \frac{n}{m} = \lim_{n\to\infty} k < \frac{2}{\delta}$.

For any $1 \le j \le m$,

$$\mathbb{P}\left(|S_j| \ge \frac{\epsilon\sigma\sqrt{m}}{9\sqrt{2\delta}}\right) = \mathbb{P}\left(\frac{|S_j|}{\sigma\sqrt{j}} \ge \frac{\epsilon\sqrt{m}}{9\sqrt{2j\delta}}\right) \le \mathbb{P}\left(\frac{|S_j|}{\sigma\sqrt{j}} \ge \frac{\epsilon}{9\sqrt{2\delta}}\right).$$

By the CLT, $\frac{|S_j|}{\sigma\sqrt{j}} \stackrel{d}{\to} Z$. Therefore

$$\mathbb{P}\left(\frac{|S_j|}{\sigma\sqrt{j}} > \frac{\epsilon}{9\sqrt{2\delta}}\right) \to \mathbb{P}\left(|Z| > \frac{\epsilon}{9\sqrt{2\delta}}\right).$$

Therefore there exists $j_0 = j_0(\epsilon, \delta)$ such that, for all $j \ge j_0$, one has

$$\mathbb{P}\left(\frac{|S_j|}{\sigma\sqrt{j}} > \frac{\epsilon}{9\sqrt{2\delta}}\right) \le 2\mathbb{P}\left(|Z| > \frac{\epsilon}{9\sqrt{2\delta}}\right) \le \frac{2 \times (9\sqrt{2})^4 \delta^2 \mathbb{E}Z^4}{\epsilon^2} = \frac{C_1\delta^2}{\epsilon^2}.$$

For $j < j_0$, we have, by Chebychev's inequality,

$$\mathbb{P}\left(\frac{|S_j|}{\sigma\sqrt{j}} \ge \frac{\epsilon\sqrt{m}}{9\sqrt{2j\delta}}\right) \le \frac{(9\sqrt{2})^2 j\delta}{\epsilon^2 m} < \frac{(9\sqrt{2})^2 j_0\delta}{\epsilon^2 m} \le \frac{C_2 j_0}{\epsilon^2 n\delta},$$

[4] This says that, for independent random variables $\xi_1, \ldots, \xi_n$, one has $\mathbb{P}(\max_{1 \le i \le n} |S_i| \ge 3\alpha) \le 3\max_{1 \le i \le n} \mathbb{P}(|S_i| \ge \alpha)$.

for some constant $C_2 > 0$. Thus we get, for all large enough n, that

$$\mathbb{P}_n(w_x(\delta) \geq \epsilon) \leq \frac{6}{\delta} \max\left\{\frac{C_1\delta^2}{\epsilon^2}, \frac{C_2 j_0}{\epsilon^2 n\delta}\right\} = \max\left\{\frac{C_3\delta}{\epsilon^2}, \frac{C_4 j_0}{\epsilon^2 n\delta^2}\right\},$$

for constants $C_3, C_4 > 0$. It follows that

$$\lim_{\delta\to 0} \limsup_{n\to\infty} \mathbb{P}_n(w_x(\delta) \geq \epsilon) = 0.$$

This verifies condition (ii) of Lemma 3.2. This proves, simultaneously, the existence of W and that $\mathbb{P}_n \xrightarrow{w} W$. □

A random function B on $C[0,1]$, whose distribution is the Wiener measure, is called a *standard Brownian motion* on $[0,1]$. Thus we have proved that $X_n \xrightarrow{d} B$.

Corollary 3.5 (Donsker). *Let $h : C[0,1] \to \mathbb{R}$ be such that $W(D_h) = 0$. Then $h(X_n) \xrightarrow{d} h(B)$.*

3.3 Some consequences of Donsker's theorem

1. Suppose $h(x) = \max_{t\in[0,1]} x(t)$. Then h is a continuous functional on $C[0,1]$, because $|h(x) - h(y)| = |\|x\|_\infty - \|y\|_\infty| \leq \|x - y\|_\infty$. Hence, we conclude that

$$\max_{1\leq i\leq n} \frac{S_i}{\sigma\sqrt{n}} = h(X_n) \xrightarrow{d} h(B),$$

whenever the ξ_i satisfy assumptions of Theorem 3.4. In particular, take ξ_1 to be a Rademacher variable: $\mathbb{P}(\xi_1 = \pm 1) = 1/2$. Then $\sigma^2 = 1$, and we have a simple symmetric random walk (SSRW), for which one can prove, by reflection principle, that

$$\mathbb{P}(\max_{1\leq i\leq n} S_i \geq a) = 2\mathbb{P}(S_n > a) + \mathbb{P}(S_n = a),$$

where a is an integer. Therefore, for $\alpha > 0$,

$$\begin{aligned}
\mathbb{P}\left(\max_{1\leq i\leq n} \frac{S_i}{\sqrt{n}} > \alpha\right) &= \mathbb{P}(\max_{1\leq i\leq n} S_i > \sqrt{n}\alpha) \\
&= \mathbb{P}(\max_{1\leq i\leq n} S_i \geq a_n) \quad (\text{where } a_n = \lceil\sqrt{n}\alpha\rceil \text{ or } \lceil\sqrt{n}\alpha\rceil + 1) \\
&= 2\mathbb{P}(S_n > a_n) + \mathbb{P}(S_n = a_n) \\
&= 2\mathbb{P}\left(\frac{S_n}{\sqrt{n}} > \frac{a_n}{\sqrt{n}}\right) + \mathbb{P}\left(\frac{S_n}{\sqrt{n}} = \frac{a_n}{\sqrt{n}}\right) \\
&\to 2\mathbb{P}(Z > \alpha).
\end{aligned}$$

(Justify this last step.) It follows that

$$\mathbb{P}\left(\max_{1\leq i\leq n} \frac{S_i}{\sqrt{n}} \leq \alpha\right) \to 1 - 2\mathbb{P}(Z > \alpha) = \mathbb{P}(|Z| \leq \alpha).$$

Thus $h(B) \stackrel{d}{=} |Z|$, which one can restate as

$$\max_{t\in[0,1]} B(t) \stackrel{d}{=} |B(1)|.$$

2. It is possible to find the joint distribution of $(m, M, B(1))$ using Donsker's theorem, where $m = \min_{t\in[0,1]} B(t), M = \max_{t\in[0,1]} B(t)$. Indeed, if $m_n = \min_{t\in[0,1]} X_n(t), M_n = \max_{t\in[0,1]} X_n(t)$, then

$$\frac{1}{\sigma\sqrt{n}}(m_n, M_n, S_n) \xrightarrow{d} (m, M, B(1)),$$

because the functional $x \mapsto (\min_{t\in[0,1]})x(t), \max_{t\in[0,1]} x(t), x(1))$ is continuous. By using properties of SSRW, one can derive the limit distribution of $\frac{1}{\sigma\sqrt{n}}(m_n, M_n, S_n)$ to show that, for $a < 0 < b, a < x < y < b$,

$$\begin{aligned}\mathbb{P}(a < m \le M < b, x < B(1) < y) \qquad (1)\\ = \sum_{k\in Z} \mathbb{P}(x + 2k(b-a) < Z < y + 2k(b-a))\\ - \sum_{k\in Z} \mathbb{P}(2b - y + 2k(b-a) < Z < 2b - x + 2k(b-a)),\end{aligned}$$

where $Z \sim \mathcal{N}(0,1)$.

3.4 The (standard) Brownian bridge

A *(standard) Brownian bridge* is a random function B^{br} in $C[0,1]$ with the property that

$$(B^{br}(t_1), \ldots, B^{br}(t_k)) \stackrel{d}{=} \mathcal{N}(0, \Sigma^{(t_1,\ldots,t_k)}),$$

where $\Sigma_{ij}^{(t_1,\ldots,t_k)} = t_i(1-t_j), i \le j$ [5]. One way to construct the standard Brownian bridge is by the following transformation on the standard Brownian motion:

$$B^{br}(t) = B(t) - tB(1).$$

We will denote the distribution of B^{br} by W^{br}. The standard Brownian bridge has a useful interpretation: it is essentially a standard Brownian motion whose value at 1 is pinned at 0. Since $\{B(1) = 0\}$ is measure zero event, care needs to be taken to define this properly.

Exercise 3.6. *Show that B^{br} and $B(1)$ are independent.*

Lemma 3.7. *For $\epsilon > 0$, define a measure on $C[0,1]$ by $\mathbb{P}_\epsilon(A) = \mathbb{P}(B \in A \mid |B(1)| \le \epsilon\}$. Then $\mathbb{P}_\epsilon \xrightarrow{w} W^{br}$ as $\epsilon \to 0$.*

Proof. We will use the facts that B^{br} and $B(1)$ are independent, and that $\|B^{br} - B\|_\infty = |B(1)|$. Let h be a bounded uniformly continuous function on $C[0,1]$. Note that

$$\begin{aligned}&|\mathbb{E}[h(B) \mid |B(1)| \le \epsilon] - \mathbb{E}[h(B^{br})]|\\ &\quad= \left|\mathbb{E}\left[h(B) \mid |B(1)| \le \epsilon\right] - \mathbb{E}\left[h(B^{br}) \mid |B(1)| \le \epsilon\right]\right|\\ &\quad\le \mathbb{E}\left[|h(B) - h(B^{br})| \mid |B(1)| \le \epsilon\right]\\ &\quad\le \mathbb{E}\left[w_h(\|B - B^{br}\|_\infty) \mid |B(1)| \le \epsilon\right]\\ &\quad\le w_h(\epsilon).\end{aligned}$$

Now let $\epsilon \to 0$. □

[5] A stochastic process X_t on $[0,1]$ is called a *Gaussian process* if all of its finite dimensional distributions are (multivariate) Gaussians. Thus both B and B^{br} are examples of Gaussian processes.

This characterization of Brownian bridge is useful for calculating limit distributions of functionals of Brownian bridge from those of Brownian motion.

As an example, we calculate the distribution of $\sup_{t\in[0,1]} |B^{br}|$. (This is an important functional, because later we will prove that it is the weak limit of the supremum of the canonical empirical process: $\sqrt{n}\sup_{t\in[0,1]} |F_n(t) - t| \xrightarrow{d} \sup_{t\in[0,1]} |B^{br}|$.)

Theorem 3.8. *Let $b > 0$. We have*

$$\mathbb{P}(\sup_{t\in[0,1]} |B^{br}| \le b) = \sum_{k\in\mathbb{Z}} (-1)^k e^{-2k^2b^2}.$$

Proof. Put $a = -b, x = -\epsilon, y = \epsilon < b$ in (1) to get

$$\begin{aligned}\mathbb{P}(-b < m \le M < b, -\epsilon < B(1) < \epsilon) \\ = \sum_{k\in Z} [\Phi(\epsilon + 4kb) - \Phi(-\epsilon + 4kb)] \\ - \sum_{k\in Z} [\Phi(\epsilon + 2(2k+1)b) - \Phi(-\epsilon + 2(2k+1)b)].\end{aligned}$$

Hence

$$\begin{aligned}\mathbb{P}(-b < m \le M < b \mid -\epsilon < B(1) < \epsilon) \\ = \sum_{k\in Z} \frac{\Phi(\epsilon + 4kb) - \Phi(-\epsilon + 4kb)}{\Phi(\epsilon) - \Phi(-\epsilon)} \\ - \sum_{k\in Z} \frac{\Phi(\epsilon + 2(2k+1)b) - \Phi(-\epsilon + 2(2k+1)b)}{\Phi(\epsilon) - \Phi(-\epsilon)}.\end{aligned}$$

Let $\epsilon \to 0$ and use Lemma 3.9 and the mapping theorem to conclude that (why is the interchange of sum and limit justified?)

$$\begin{aligned}\mathbb{P}(-b < m^{br} \le M^{br} < b) &= \sum_{k\in Z} \frac{\phi(4kb)}{\phi(0)} - \sum_{k\in Z} \frac{\phi(2(2k+1)b)}{\phi(0)} \\ &= \sum_{k\in Z} e^{-2(2k)^2b^2} - \sum_{k\in Z} e^{-2(2k+1)^2b^2} \\ &= \sum_{k\in\mathbb{Z}} (-1)^k e^{-2k^2b^2}.\end{aligned}$$

As $\mathbb{P}(\sup_{t\in[0,1]} |B^{br}| \le b) = \mathbb{P}(-b \le m^{br} \le M^{br} \le b)$, we are done. □

3.5 Brownian motion on $[0,\infty)$

Using the standard Brownian bridge, we may construct standard Brownian motion on $[0,\infty)$ via the following transformation:

$$\tilde{B}(t) = (1+t)B^{br}\left(\frac{t}{1+t}\right), t \ge 0.$$

Exercise 3.9. *Show that for $s, t \in [0,\infty)$, $\mathrm{cov}(\tilde{B}(s), \tilde{B(t)}) = s \wedge t$.*

Question: What would be the right topology on $C[0,\infty)$ for this to make sense? That is, if Ψ is the map taking $x \in C[0,1]$ to $\Psi(x)(t) = (1+t)x\left(\frac{t}{1+t}\right) \in C[0,\infty)$, then what topology on $C[0,\infty)$ would make this map Borel measurable, so that one can define Wiener measure on $C[0,\infty)$ as $W^{br} \circ \Psi^{-1}$?

It turns out that the right topology should be the *topology of uniform convergence on compact sets*. This is metrizable, e.g., using the following metric

$$\rho(x,y) = \sum_{k\geq 1} \frac{1}{2^k} \min\{ \sup_{t\in[0,k]} |x(t) - y(t)|, 1\}.$$

Exercise 3.10. *Show that $C[0,\infty)$, equipped with ρ, is a Polish space.*

Exercise 3.11. *Ψ is continuous. In fact, $\rho(\Psi(x),\Psi(y)) \leq C\|x-y\|_\infty$, for some universal constant $C > 0$.*

Thus we can define the Wiener measure on $C[0,\infty)$ as the pushforward $W^{br} \circ \Psi^{-1}$ of W^{br} via the map Ψ.

An alternative route is to characterize tightness on $(C[0,\infty),\rho)$ using a suitable Arzelà-Ascoli theorem and then prove weak convergence of the following random functions

$$\tilde{X}_n(t) = \frac{S_{\lfloor nt \rfloor}}{\sigma\sqrt{n}} + (nt - \lfloor nt \rfloor)\frac{\xi_{\lfloor nt \rfloor + 1}}{\sigma\sqrt{n}}, t \geq 0.$$

The limit, naturally, would be Brownian motion on $C[0,\infty)$. A treatment of this way of constructing $\tilde{B}$ can be found in Section 2.4 of Karatzas and Shreve (2012).

4 Weak convergence in $D[0,1]$

$D[0,1]$ or D for short is the set of functions on $[0,1]$ that are right continuous and have left limits. These functions are often called *càdlàg* (abbreviation for the french term *continue à droite, limite à gauche*), or RCLL (abbreviation for Right Continuous with Left Limits), or CORLOL (abbreviation for Continuous On the Right, Limit On the Left) functions. Examples are $\mathbf{I}_{[0,a]}(t)$ or $\mathbf{I}_{[a,1]}(t)$. Obviously, $C[0,1] \subset D[0,1]$. $D[0,1]$ is an importance space to develop weak convergence theory for. Many natural stochastic processes have discontinuous paths living in $D[0,1]$. For example, various jump processes such as the Poisson process, the empirical process and so on. Moreover, the random walk process

$$X_n(t) = \frac{1}{\sigma\sqrt{n}} S_{\lfloor nt \rfloor}$$

has sample paths in $D[0,1]$ (and converges to Brownian motion, as we shall establish in this section).

4.1 The Skorohod topology

We need a suitable topology on $D[0,1]$. The uniform metric $\|\cdot\|_\infty$ is too strong to be useful. For example, it fails to capture the closeness of the functions $\mathbf{I}_{[0,a)}(t)$ and $\mathbf{I}_{[0,a+\frac{1}{n})}(t)$. Before we introduce a suitable metric on $D[0,1]$, let us introduce an analogue of the modulus of continuity $w_x(\delta)$ for $D[0,1]$, which will help us express results in a succinct way.

The variation of a function x over a set $A \subset [0,1]$ is defined as

$$w_x(A) = \sup_{u,v \in A} |x(u) - x(v)|.$$

In this notation,

$$w_x(\delta) = \sup_{t \in [0,1-\delta]} w_x([t, t+\delta]).$$

Since functions in $D[0,1]$ may have jumps, $w_x(\delta)$ may be arbitarily large! The problem is that we cannot allow every t in the supremum in the definition of $w_x(\delta)$. We have to avoid the jump points.

Exercise 4.1. *Construct a sequence of functions $x_n \in D$ such that $\liminf_n w_{x_n}(\delta) = \infty$, for any $\delta > 0$.*

Recall that, a δ-sparse partition $\sigma = \{t_i\}$ of $[0,1]$ is a set of points $0 = t_0 < t_1 < \cdots < t_k = 1$ such that $\min_i\{t_i - t_{i-1}\} \geq \delta$. Define

$$w'_x(\delta) := \inf_{\{t_i\}\ \delta\text{-sparse}} \max_i w_x[t_{i-1}, t_i).$$

Exercise 4.2. *Show that $w'_x(\delta)$ is a nondecreasing function of δ.*

Lemma 4.3. *Let $\epsilon > 0$. There exists a δ-sparse partition $\{t_i\}$ such that*

$$\max_i w_x[t_{i-1}, t_i) < \epsilon.$$

This and the above lemma implies that

$$x \in D \text{ if and only if } \lim_{\delta \to 0} w'_x(\delta) = 0.$$

Now the above lemma also implies that any $x \in D$ has at most countably many jump discontinuities and that $\|x\|_\infty < \infty$.

Given a partition $\sigma = \{t_i\}$, consider the map $S_\sigma : D[0,1] \to D[0,1]$, taking x to the simple function

$$S_\sigma x(t) = \sum_i x(t_{i-1}) \mathbf{I}_{[t_{i-1},t_i)}(t) + x(1)\mathbf{I}_{\{1\}}(t).$$

It is clear that

$$\|x - S_\sigma x\|_\infty \leq \max_i w_x[t_{i-1}, t_i).$$

Therefore, one can find a sequence of partitions σ_n such that $S_{\sigma_n} x \to x$ uniformly. Thus every càdlàg function is Borel measurable.

Let Λ be the group of continuous strictly increasing surjective functions λ on $[0,1]$. Clearly, $\lambda(0) = 0$, $\lambda(1) = 1$. We will think of λ's as deformations of the time scale. Going back to the previous example, we want to devise a metric that not only compares the values of $\mathbf{I}_{[0,a+\frac{1}{n})}(t)$ and $\mathbf{I}_{[0,a)}(t)$, but also their supports. λ's will help us accomplish the latter. Define

$$d(x,y) = \inf_{\lambda \in \Lambda} \|x - y \circ \lambda\|_\infty \vee \|\lambda - I\|_\infty.$$

Exercise 4.4. *Check that Λ indeed is a group.*

5 More on Empirical Processes

In the last section we proved that the unifrom empirical process $U_n(t) = \sqrt{n}\|F_n(t) - t\|$ converges weakly to the standard Brownian bridge (in the Skorohod topology). This is a uniform analog of the CLT. We will now focus on proving uniform laws for more general empirical processes. Let $\mathcal{F}$ be a class of measurable functions. Let $\xi_i \overset{i.i.d.}{\sim} \mathbb{P}$. The *empirical process indexed by $\mathcal{F}$* is the stochastic process

$$(\mathbb{P}_n - \mathbb{P})f = \mathbb{P}_n f - \mathbb{P}f = \frac{1}{n}\sum_{i=1}^{n} f(\xi_i) - \mathbb{P}f.$$

If we take $\mathcal{F} = \{\mathbf{1}_{(-\infty,t]}(\cdot) \mid t \in \mathbb{R}\}$, then we recover the uniform empirical process studied earlier.

A classical result in statistics is the Glivenko-Cantelli theorem:

$$\sup_{t\in\mathbb{R}} |F_n(t) - F(t)| \overset{a.s.}{\to} 0,$$

which can be thought of as a uniform law of large numbers (because SLLN only guarantees that $\hat{F}_n(t) - F(t) \overset{a.s.}{\to} 0$, for each fixed t).

In this section, we will develop methods to prove such uniform laws for more general empirical processes.

To this end, call a class $\mathcal{F}$ of measurable functions *$\mathbb{P}$-Glivenko-Cantelli* if

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} := \sup_{f\in\mathcal{F}} |\mathbb{P}_n f - \mathbb{P}f| \overset{a.s.}{\to} 0.$$

Because of the classical Glivenko-Cantelli theorem, $\mathcal{F} = \{\mathbf{1}_{(-\infty,t]}(\cdot) \mid t \in \mathbb{R}\}$ is a Glivenko-Cantelli class.

Remark 5.1 (Measurability issues)**.** *There is an obvious question whether $\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}}$ is measurable. There are some standard techniques to deal with this issue such as (a) the outer-expectation approach of Hoffman-Jorgensen, (b) working with separable stochastic processes only, etc. We will henceforth not address measurability issues at all. If you are uncomfortable, assume that the function class $\mathcal{F}$ is countable—no measurability issues would arise then.*

Exercise 5.1. *Suppose that $\mathbb{P}$ is a discrete probability measure. Then $\mathcal{F} = \{\mathbf{1}_A(\cdot) \mid A \text{ is a Borel set}\}$ is $\mathbb{P}$-Glivenko-Cantelli.*

Exercise 5.2. *Show that the above statement is false for absolutely continuous probability measures on $\mathbb{R}$.*

One motivation behind proving uniform laws is to establish consistency of estimators. For example, the median of a univariate distribution F defined as

$$\text{med}(F) = \inf\{x \mid F(x) \geq 1/2\}$$

is a continous functional in the Kolmogorov-Smirnov metric $\delta(F, G) = \sup_{t\in\mathbb{R}} |F(t) - G(t)|$. Thus, by the Glivenko-Cantelli theorem, the sample median $\text{med}(\hat{F}_n)$ is a consistent estimator for the population median.

5.1 Concentration inequalities

In this section, we will prove two useful concentration inequalities in a series of exercises: Hoeffding's inequality and the bounded-differences inequality.

A random variable X with mean μ is called *sub-Gaussian with parameter σ* if

$$\mathbb{E}e^{\lambda(X-\mu)} \leq e^{\frac{\sigma^2\lambda^2}{2}} \text{ for all } \lambda \in \mathbb{R}.$$

This is essentially saying that the tails of X as fast as a Gaussian.

Exercise 5.3. *Suppose X is sub-Gaussian with parameter σ. Then*

$$\mathbb{P}(X \geq \mathbb{E}X + \delta) \leq e^{-\frac{\delta^2}{2\sigma^2}}.$$

Exercise 5.4. *Suppose X_1 and X_2 are independent sub-Gaussians with parameters σ_1 and σ_2 respectively. Then $a_1X_1 + a_2X_2$ is sub-Gaussian with parameter $\sqrt{a_1^2\sigma_1^2 + a_2^2\sigma_2^2}$. (Note: Independence is not actually necessary, but then you would get a slightly worse sub-Gaussianity parameter.)*

Exercise 5.5 (Hoeffding's inequality)**.** *Suppose X_i are independent sub-Gaussians with parameters $\sigma_i, i = 1, \ldots, n$. Then*

$$\mathbb{P}(\sum_{i=1}^n a_iX_i \geq \mathbb{E}\sum_{i=1}^n a_iX_i + \delta) \leq e^{-\frac{\delta^2}{2\sum_{i=1}^n a_i^2\sigma_i^2}}.$$

Exercise 5.6. *Suppose X_i are zero-mean independent sub-Gaussians with common parameter $\sigma, i = 1, \ldots, n$. Then*

$$\mathbb{P}(\frac{1}{n}\sum_{i=1}^n X_i \geq \delta) \leq e^{-\frac{n\delta^2}{2\sigma^2}}.$$

Exercise 5.7. *Suppose η takes values ± 1 with probability $1/2$ each (such random variables are called Rademacher variables). Show that η is sub-Gaussian with parameter 1.*

Exercise 5.8. *Let X be a random variable such that $a \leq X \leq b$ almost surely. Show that X is sub-Gaussian with parameter $\sigma = (b-a)/2$. (Hint: Consider Taylor-expanding the cumulant generating function $\psi(\lambda) = \log \mathbb{E}e^{\lambda X}$ of X around 0.)*

Using a soft symmetrization argument one can show that bounded random variables are sub-Gaussian with parameter $(b-a)$. Here is how it goes. Let X' be an independent copy of X and let η be a Rademacher variable independet of both X and X'. We have

$$\mathbb{E}_X e^{\lambda(X-\mathbb{E}X)} = \mathbb{E}_X e^{\lambda(X-\mathbb{E}X')} = \mathbb{E}_X e^{\mathbb{E}_{X'}\lambda(X-X')} \overset{\text{(Jensen)}}{\leq} \mathbb{E}_X\mathbb{E}_{X'}e^{\lambda(X-X')}.$$

Now $\eta(X-X') \overset{d}{=} (X-X')$. Hence

$$\mathbb{E}_X\mathbb{E}_{X'}e^{\lambda(X-X')} = \mathbb{E}_X\mathbb{E}_{X'}\mathbb{E}_\eta e^{\lambda\eta(X-X')} \leq \mathbb{E}_X\mathbb{E}_{X'}e^{\frac{\lambda^2(X-X')^2}{2}} \leq e^{\frac{\lambda^2(b-a)^2}{2}}.$$

Expected maxima of sub-Gaussians will appear many times in the later subsections.

Exercise 5.9. *Let $X_1, \dots, X_n$ be zero-mean sub-Gaussians with common parameter σ. Show that*

$$\mathbb{E} \max_{1 \le i \le n} X_i \le \sqrt{2\sigma^2 \log n}$$

for all $n \ge 1$. If one replaces X_i with $|X_i|$, show then that an upper bound of $2\sqrt{\sigma^2 \log n}$ holds for all $n \ge 2$.

We will now go beyond independence. Let $(Y_k)_{k \ge 0}$ be a martingale with respect to some filtration $(\mathcal{F}_k)_{k \ge 0}$. Let $D_k = Y_k - Y_{k-1}, k \ge 1$ be the corresponding sequence of martingale-differences. Assume that

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \le e^{\frac{\lambda^2 \sigma_k^2}{2}}.$$

(As pointed out by Apratim, this condition implies, via Exercise 1 of Homework 5, that D_k is a sequence of martingale-differences.)

Exercise 5.10. *Show that under the above setup we have*

$$\mathbb{P}(\sum_{k=1}^n D_k \ge \delta) \le e^{-\frac{\delta^2}{2\sum_{k=1}^n \sigma_k^2}}.$$

Exercise 5.11 (Azuma-Hoeffding inequality)**.** *Suppose that the martingale-difference $D_k \in [a_k, b_k]$ a.s.*

(a) Show that

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \le e^{\frac{\lambda^2 (b_k - a_k)^2}{8}}.$$

(b) Conclude that

$$\mathbb{P}(\sum_{k=1}^n D_k \ge \delta) \le e^{-\frac{2\delta^2}{\sum_{k=1}^n (b_k - a_k)^2}}.$$

A standard way to construct martingales is the Doob-construction: Given a random vector $X = (X_1, \dots, X_n)$ with *independent components* and some function $f : \mathbb{R}^n \to \mathbb{R}$ such that $\mathbb{E}|f(X)| < \infty$, let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$ and

$$Y_k = \mathbb{E}[f(X) \mid \mathcal{F}_k].$$

Indeed, by the tower property of conditional expectations,

$$\mathbb{E}[Y_k \mid \mathcal{F}_{k-1}] = \mathbb{E}[[f(X) \mid \mathcal{F}_k] \mid \mathcal{F}_{k-1}] = \mathbb{E}[f(X) \mid \mathcal{F}_{k-1}] = Y_{k-1}.$$

Exercise 5.12 (Bounded-differences inequality)**.** *Suppose that the function f has the bounded-differences property, i.e., for any $1 \le i \le n$, there is some positive constant b_i such that*

$$|f(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_{i-1}, x_i', x_{i+1}, \dots, x_n)| \le \ell_i$$

for all $x_i, x_i' \in \mathbb{R}$.

(a) Show that the corresponding Doob-martingale-difference D_k satisfies

$$a_k \le D_k \le b_k \text{ a.e. for some } a_k \le b_k \text{ with } b_k - a_k \le \ell_k.$$

(b) *Hence establish the* bounded-differences inequality

$$\mathbb{P}(f(X) \geq \mathbb{E}f(X) + \delta) \leq e^{-\frac{2\delta^2}{\sum_{k=1}^n \ell_k^2}}.$$

The bounded differences inequality may be used to show concentration of (a) U-statistics, (b) length of the longest common subsequence of two independent random equal-length finite sequences from a common alphabet, (c) Rademacher complexity, and so on.

5.2 Covering number and bracketing numbers

Whether or not a class $\mathcal{F}$ is Glivenko-Cantelli will depend on the "size" of the class. Let us discuss some notions of size. Let T be a totally bounded set in a metric space $(\mathcal{M}, \rho)$. The *ϵ-covering number* $N(\epsilon, T, \rho)$ is the minimum size of an ϵ-net for T. The natural logarithm of $N(\epsilon, T, \rho)$ is sometimes called *metric entropy* for statistical mechanical reasons.

Exercise 5.13. *Show that the ϵ-covering number of a d-dimensional Euclidean ball satisfies*

$$N(\epsilon, B(x; R), \|\cdot\|_2) \leq \left(1 + \frac{2}{\epsilon}\right)^d.$$

Now let $\mathcal{M}$ be a pseudo-metric space of functions. The set $[\ell, u] = \{f \in \mathcal{M} \mid \ell \leq f \leq u, \rho(u, \ell) < \epsilon\}$ is called an *ϵ-bracket*. The *ϵ-bracketing number* $N_{[]}(\epsilon, T, \rho)$ is the minimum size of a cover of T by ϵ-brackets.

The classical Glivenko-Cantelli theorem may be rephrased in terms of ϵ-bracketing numbers in the $L_1(\mathbb{P})$ (semi-)norm.

Theorem 5.14. *Let $\mathcal{F}$ be a class of measurable functions such that $N_{[]}(\epsilon, \mathcal{F}, \|\cdot\|_{L_1}) < \infty$ for any $\epsilon > 0$. Then $\mathcal{F}$ is $\mathbb{P}$-Glivenko-Cantelli.*

Proof. Choose a covering $[\ell_i, u_i], i = 1, \ldots, k$ by ϵ-brackets. Given $f \in \mathcal{F}$ find a bracket $[\ell_j, u_j]$ enveloping f. Then note that

$$\begin{aligned}
\mathbb{P}_n f - \mathbb{P} f &\leq \mathbb{P}_n u_j - \mathbb{P} f \\
&= (\mathbb{P}_n - \mathbb{P}) u_j + \mathbb{P}(u_j - f) \\
&\leq \max_j (\mathbb{P}_n - \mathbb{P}) u_j + \epsilon.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{P} f - \mathbb{P}_n f &\leq \mathbb{P} f - \mathbb{P}_n \ell_j \\
&= -(\mathbb{P}_n - \mathbb{P}) \ell_j + \mathbb{P}(f - \ell_j) \\
&\leq -\min_j (\mathbb{P}_n - \mathbb{P}) \ell_j + \epsilon.
\end{aligned}$$

Thus

$$|\mathbb{P}_n f - \mathbb{P} f| \leq \underbrace{\max\{\max_j (\mathbb{P}_n - \mathbb{P}) u_j, -\min_j (\mathbb{P}_n - \mathbb{P}) \ell_j\}}_{=:\Delta_{n,\epsilon}} + \epsilon.$$

The right hand side does not depend on f, and hence serves as a unifrom upper bound

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \leq \Delta_{n,\epsilon} + \epsilon.$$

By SLLN $(\mathbb{P}_n - \mathbb{P})u_j, (\mathbb{P}_n - \mathbb{P})\ell_j \overset{a.s.}{\to} 0$, and since we have finitely many functions u_j, ℓ_j, we conclude that $\Delta_{n,\epsilon} \overset{a.s.}{\to} 0$.

Thus, for each fixed $\epsilon > 0$,

$$\limsup_{n\to\infty} \|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \leq \epsilon \text{ a.s.}$$

Choosing a sequence $\epsilon_k \to 0$ and adjusting the null set accordingly, we conclude that

$$\limsup_{n\to\infty} \|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} = 0 \text{ a.s.}$$

This completes the proof. □

Exercise 5.15. *Recover the classical Glivenko-Cantelli theorem by exhibiting a cover of* $\mathcal{F} = \{\mathbf{1}_{(-\infty,t]}(\cdot) \mid t \in \mathbb{R}\}$ *by* ϵ*-brackets for each* $\epsilon > 0$.

We will now discuss a very useful trick called *symmetrization*, which, coupled with a size bound will yield a host of Glivenko-Cantelli theorems.

5.3 Symmetrization

Recall that a Rademacher random variable η takes values ± 1 with equal probability. Take an i.i.d. vector of such variables $\eta = (\eta_i, \ldots, \eta_d)$. Clearly, η is a uniformly chosen corner of the d-dimensional hypercube of side 2 centered at the origin. The size (or, perhaps more accurately "spread") of a set $A \subset \mathbb{R}^d$ can be measured by the maximum inner product of an element of A with such a random Rademacher direction. We define the *Rademacher complexity* of A as

$$\mathcal{R}(A) := \mathbb{E}_\eta \sup_{a\in A} \langle a, \eta \rangle.$$

Given a function class $\mathcal{F}$ and n i.i.d. random elements ξ_i, the average size (w.r.t. the distribution of the ξ_j's) of $\mathcal{F}$ may be measured by the expected Rademacher complexity of the Euclidean set $\mathcal{F}(\xi) := \{\frac{1}{n}(f(\xi_1), \ldots, f(\xi_n)) \mid f \in \mathcal{F}\}$ (intuitively, if $\mathcal{F}$ is large, one would expect the Euclidean set $\mathcal{F}(\xi)$ to be large as well). We define this to be the Rademacher complexity of $\mathcal{F}$:

$$\mathcal{R}(\mathcal{F}) := \mathbb{E}_\xi \mathcal{R}(\mathcal{F}(\xi)).$$

Theorem 5.16 (symmetrization)**.** *We have*

$$\mathbb{E}\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \leq 2\mathcal{R}(\mathcal{F}).$$

Proof. Let $\xi_i', i = 1, \ldots, n$ be independent copies of $\xi_i, i = 1, \ldots, n$. Using Jensen, we have

$$\begin{aligned}
\mathbb{E}\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} &= \mathbb{E}_{\xi} \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} f(\xi_i) - \mathbb{P}(f)| \\
&= \mathbb{E}_{\xi} \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} (f(\xi_i) - \mathbb{E}_{\xi_i'} f(\xi_i'))| \\
&= \mathbb{E}_{\xi} \sup_{f \in \mathcal{F}} |\mathbb{E}_{\xi'} \frac{1}{n} \sum_{i=1}^{n} (f(\xi_i) - f(\xi_i'))| \\
&\leq \mathbb{E}_{\xi} \sup_{f \in \mathcal{F}} \mathbb{E}_{\xi'} |\frac{1}{n} \sum_{i=1}^{n} (f(\xi_i) - f(\xi_i'))| \\
&\leq \mathbb{E}_{\xi} \mathbb{E}_{\xi'} \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} (f(\xi_i) - f(\xi_i'))|.
\end{aligned}$$

Now let η_i be i.i.d. Rademacher, independent of the ξ_i's and the ξ_i''s. Then

$$(\eta_i(f(\xi_i) - f(\xi_i')))_{1 \leq i \leq n} \stackrel{d}{=} (f(\xi_i) - f(\xi_i'))_{1 \leq i \leq n}.$$

Ignoring measurability issues, we conclude that

$$\sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} (f(\xi_i) - f(\xi_i'))| \stackrel{d}{=} \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} \eta_i (f(\xi_i) - f(\xi_i'))|.$$

Therefore

$$\begin{aligned}
\mathbb{E}_{\xi} \mathbb{E}_{\xi'} \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} (f(\xi_i) - f(\xi_i'))| &= \mathbb{E}_{\xi} \mathbb{E}_{\xi'} \mathbb{E}_{\eta} \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} \eta_i (f(\xi_i) - f(\xi_i'))| \\
&\leq \mathbb{E}_{\xi} \mathbb{E}_{\xi'} \mathbb{E}_{\eta} \left[\sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} \eta_i f(\xi_i)| + \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^{n} \eta_i f(\xi_i)| \right] \\
&= 2\mathcal{R}(\mathcal{F}).
\end{aligned}$$

This concludes the proof. □

We end this section with an application of the bounded-differences inequality.

Exercise 5.17. *Let $\mathcal{F}$ be a class of functions that are uniformly bounded by B. Show that*

$$\mathbb{P}(\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \geq \mathbb{E}\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} + \delta) \leq e^{-\frac{n\delta^2}{2B^2}}.$$

Conclude that, with probability at least $1 - e^{-\frac{n\delta^2}{2B^2}}$, we have

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} < \mathbb{E}\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} + \delta \leq 2\mathcal{R}(\mathcal{F}) + \delta.$$

In the next subsection, we will learn how to bound the Rademacher complexity $\mathcal{R}(\mathcal{F})$ of uniformly bounded function classes of finite VC-dimension.

5.4 Discretization and chaining

We saw objects like $\mathbb{E}\sup_{t\in\mathbb{T}} X_t$ appear frequently in the previous subsection. For example, in the definition of Rademacher complexity. In this subsection we will discuss a technique for (upper)-bounding these objects.

A zero-mean stochastic process $(X_t)_{t\in\mathbb{T}}$, indexed by a (pseudo-)metric space $(\mathbb{T},\rho)$ is called *sub-Gaussian w.r.t. ρ* if

$$\mathbb{E}e^{\lambda(X_t-X_{t'})} \le e^{\frac{\lambda^2\rho^2(t,t')}{2}},$$

for all $t,t'\in\mathbb{T}$.

Example 5.18. *Let $\mathbb{T}=\mathcal{F}$, a class of measurable functions (which we will later assume to be uniformly bounded and have finite VC dimension). Let*

$$X_f = \frac{1}{\sqrt{n}}\sum_{i=1}^n \eta_i f(\xi_i),$$

where η_i are i.i.d. Rademacher and ξ_i's are fixed. One can then check that $(X_f)_{f\in\mathcal{F}}$ is sub-Gaussian w.r.t. the pseudo-metric

$$\rho(f,g) = \|f-g\|_{\mathbb{P}_n} = \sqrt{\frac{1}{n}\sum_{i=1}^n (f(\xi_i)-g(\xi_i))^2}.$$

Example 5.19. *Let $\mathbb{T}=\mathbb{R}^{n\times p}$. Let X be a random $n\times p$ matrix with independent zero-mean sub-Gaussian entries with sub-Gaussianity parameter 1. Let $X_\Theta = \langle X,\Theta\rangle_F = \mathrm{trace}(X\Theta^\top)$. Then it is easy to check that $(X_\Theta)_{\Theta\in\mathbb{T}}$ is sub-Gaussian w.r.t. the Frobenius metric*

$$\rho(\Theta,\Theta') = \|\Theta-\Theta'\|_F.$$

Now we prove a preliminary result on upper bounding $\sup_{t\in T} X_t$ for a sub-Gaussian process $(X_t)_{t\in\mathbb{T}}$.

Theorem 5.20 (Crude discretization)**.** *Let $(X_t)_{t\in T}$ be a sub-Gaussian stochastic process with respect to a pseudo-metric ρ on $\mathbb{T}$ under which $\mathbb{T}$ is totally bounded.*

$$\mathbb{E}\sup_{t\in\mathbb{T}} X_t \le \mathbb{E}\sup_{t,t'\in T}(X_t-X_{t'}) \le 2\mathbb{E}\sup_{\substack{t,t'\in\mathbb{T},\\ \rho(t,t')\le\delta}}(X_t-X_{t'}) + 4D\sqrt{\log N(\delta,\mathbb{T},\rho)},$$

where $D=\sup_{t,t'\in\mathbb{T}}\rho(t,t')$ is the diameter of $\mathbb{T}$.

Proof. The first inequality holds because our process is zero-mean:

$$\begin{aligned}\mathbb{E}\sup_{t\in\mathbb{T}} X_t &= \mathbb{E}\sup_{t\in\mathbb{T}}(X_t - X_{t_0}) \quad (\text{for some } t_0\in\mathbb{T})\\ &\le \mathbb{E}\sup_{t,t'\in\mathbb{T}}(X_t-X_{t'}).\end{aligned}$$

To prove the second inequality, let $N = N(\delta, \mathbb{T}, \rho)$ and let $S = \{t_1, \ldots, t_N\}$ be a δ-net of $\mathbb{T}$. Then, for any t, find $t_j \in S$ which is δ-close to t. We have

$$\begin{aligned}\pm(X_t - X_{t_1}) &= \pm(X_t - X_{t_j}) \pm (X_{t_j} - X_{t_1}) \\ &\leq \sup_{\substack{t,t' \in \mathbb{T}, \\ \rho(t,t') \leq \delta}} (X_t - X_{t'}) + \max_j |X_{t_j} - X_{t_1}|.\end{aligned}$$

Therefore

$$\begin{aligned}X_t - X_{t'} &= X_t - X_{t_1} - (X_{t'} - X_{t_1}) \\ &\leq 2 \sup_{\substack{t,t' \in \mathbb{T}, \\ \rho(t,t') \leq \delta}} (X_t - X_{t'}) + 2 \max_j |X_{t_j} - X_{t_1}|.\end{aligned}$$

The above bound, being uniform in t, t' also works for $\sup_{t,t' \in \mathbb{T}}(X_t - X_{t'})$. Taking expectations, we get

$$\mathbb{E} \sup_{t,t' \in T} (X_t - X_{t'}) \leq 2\mathbb{E} \sup_{\substack{t,t' \in \mathbb{T}, \\ \rho(t,t') \leq \delta}} (X_t - X_{t'}) + 2\mathbb{E} \max_j |X_{t_j} - X_{t_1}|.$$

Now, because $(X_t)_{t \in \mathbb{T}}$ is sub-Gaussian w.r.t. ρ, the random variable $X_{t_j} - X_{t_1}$ is sub-Gaussian w.r.t. parameter $\rho^2(t_j, t_1) \leq D^2$. Hence, by Exercise 5.9, we get

$$\mathbb{E} \max_j |X_{t_j} - X_{t_1}| \leq 2\sqrt{D^2 \log N(\delta, \mathbb{T}, \rho)}.$$

This completes the proof. □

Example 5.21. *Consider upper bounding $\mathbb{E}_\eta \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^n \eta_i f(\xi_i)|$. This appears in the definition of Rademacher complexity of a function class $\mathcal{F}$—for fixed ξ_i's we take expectation w.r.t. the η_i's. Assume that $\mathcal{F}$ is uniformly bounded (by B, say) and that is has finite VC-dimension ν.*

As we saw before, $X_f = \frac{1}{\sqrt{n}} \sum_{i=1}^n \eta_i f(\xi_i)$ is sub-Gaussian w.r.t. the pseudo-metric $\|f - g\|_{\mathbb{P}_n}^2$. Therefore, by the crude discretization bound, we have

$$\mathbb{E} \sup_{f \in \mathcal{F}} X_f \leq 2\mathbb{E} \sup_{\substack{f,g \in \mathcal{F}, \\ \|f-g\|_{\mathbb{P}_n} \leq \delta}} (X_f - X_g) + 4D\sqrt{\log N(\delta, \mathcal{F}, \|\cdot\|_{\mathbb{P}_n})}$$

Now, $X_f - X_g = \frac{1}{\sqrt{n}} \sum_{i=1}^n \eta_i (f(\xi_i) - g(\xi_i)) \overset{(\mathrm{C-S})}{\leq} \|\eta\|_2 \|f - g\|_{\mathbb{P}_n}$, whence

$$\mathbb{E} \sup_{\substack{f,g \in \mathcal{F}, \\ \|f-g\|_{\mathbb{P}_n} \leq \delta}} (X_f - X_g) \leq \delta \mathbb{E}\|\eta\|_2 \leq \delta\sqrt{\mathbb{E}\|\eta\|_2^2} = \delta\sqrt{n}.$$

It can be shown that

$$N(\delta, \mathcal{F}, \|\cdot\|_{\mathbb{P}_n}) \leq C_{B,\nu} \delta^{-2(\nu-1)},$$

for some constant $C_{B,\nu}$ that only depends on B and ν. Also note that $D \leq 2B$. Therefore

$$\mathbb{E} \sup_{f \in \mathcal{F}} X_f \leq 2\delta\sqrt{n} + 8B\sqrt{\log C_{B,\nu} + 2(\nu - 1)\log\left(\frac{1}{\delta}\right)}$$

Plugging in $\delta = \frac{1}{\sqrt{n}}$*, we get that*

$$\mathbb{E} \sup_{f \in \mathcal{F}} X_f \leq C'_{B,\nu} \sqrt{\log n},$$

for some constant $C'_{B,\nu}$ *that only depends on* B *and* ν*. Coming back to the original object of consideration,*

$$\begin{aligned}
\mathbb{E}_\eta \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^n \eta_i f(\xi_i)| &= \frac{1}{\sqrt{n}} \mathbb{E} \sup_{f \in \mathcal{F}} |X_f| \\
&\leq \frac{2}{\sqrt{n}} \mathbb{E} \sup_{f \in \mathcal{F}} X_f \\
&\leq 2C'_{B,\nu} \sqrt{\frac{\log n}{n}},
\end{aligned}$$

where we have used that fact that $X_f \overset{d}{=} -X_f$ *so that*

$$\begin{aligned}
\mathbb{E} \sup_{f \in \mathcal{F}} |X_f| &= \mathbb{E} \sup_{f \in \mathcal{F}} X_f \vee (-X_f) \\
&\leq \mathbb{E} \sup_{f \in \mathcal{F}} X_f \vee \sup_{f \in \mathcal{F}} X_f(-X_f) \\
&\leq \mathbb{E}(\sup_{f \in \mathcal{F}} X_f + \sup_{f \in \mathcal{F}} X_f(-X_f)) \\
&= 2\mathbb{E} \sup_{f \in \mathcal{F}} X_f.
\end{aligned}$$

In the bound

$$\mathbb{E}_\eta \sup_{f \in \mathcal{F}} |\frac{1}{n} \sum_{i=1}^n \eta_i f(\xi_i)| = O_{B,\nu}\left(\sqrt{\frac{\log n}{n}}\right),$$

the $\log n$ *factor is an artifact of the crudeness of our discretization and can be removed via a much more refined way of discretization called chaining.*

Example 5.22. *Consider upper bounding* $\mathbb{E}\frac{1}{\sqrt{n}}\|X\|_{\text{op}}$[6] *where* X *is as in Example 5.8. We have, using the variational representation of the operator norm,*

$$\begin{aligned}
\|X\|_{\text{op}} &= \sup_{\|x\|_2 = \|y\|_2 = 1} x^\top X y \\
&= \sup_{\|x\|_2 = \|y\|_2 = 1} \text{trace}(X y x^\top) \\
&= \sup_{\|x\|_2 = \|y\|_2 = 1} \langle X, x y^\top \rangle_F \\
&\leq \sup_{\Theta \in \mathbb{T}} \langle X, \Theta \rangle_F,
\end{aligned}$$

[6]It can be shown that the largest eigenvalue of the scaled Wishart matrix $\frac{1}{n}X^\top X$ converges almost surely to $(1+\sqrt{\gamma})^2$ in the asymptotic regime $p/n \to \gamma > 0$. Here we are using a cheap discretization bound to prove a non-asymptotic version of this.

where $\mathbb{T} = \{\Theta \in \mathbb{R}^{n\times p} \mid \operatorname{rank}(\Theta) = 1, \|\Theta\|_F = 1\}$. *Consider the stochastic process* $X_\Theta = \langle X, \Theta\rangle_F$. *From Example 5.8, we know that it is sub-Gaussian w.r.t. the Frobenius norm. Therefore, by the crude discretization bound, we get that*

$$\mathbb{E}\|X\|_{\mathrm{op}} \le 2\mathbb{E} \sup_{\substack{\Theta,\Theta'\in\mathbb{T},\\ \|\Theta-\Theta'\|_F\le\delta}} (X_\Theta - X_{\Theta'}) + 4D\sqrt{\log N(\delta, \mathbb{T}, \|\cdot\|_F)}.$$

From Exercise 5.24,

$$X_\Theta - X_{\Theta'} = \langle X, \Theta - \Theta'\rangle_F \le 2\|\Theta - \Theta'\|_F \|X\|_{\mathrm{op}}.$$

Therefore

$$\mathbb{E} \sup_{\substack{\Theta,\Theta'\in\mathbb{T},\\ \|\Theta-\Theta'\|_F\le\delta}} (X_\Theta - X_{\Theta'}) \le 2\delta\mathbb{E}\|X\|_{\mathrm{op}}.$$

Using Exercise 5.13, it is easy to show that

$$N(\delta, \mathbb{T}, \|\cdot\|_F) \le \left(1 + \frac{2}{\delta}\right)^{n+p}.$$

Also, note that $D \le 2$. *Therefore*

$$\mathbb{E}\|X\|_{\mathrm{op}} \le 4\delta\mathbb{E}\|X\|_{\mathrm{op}} + 8\sqrt{(n+p)\log\left(1 + \frac{2}{\delta}\right)}.$$

Re-arranging and choosing $\delta = \delta_0$, *a constant, such that* $1 - 4\delta_0 > 0$, *we get*

$$\mathbb{E}\|X\|_{\mathrm{op}} \le C\sqrt{n+p},$$

for some universal constant C. *Hence*

$$\mathbb{E}\frac{1}{\sqrt{n}}\|X\|_{\mathrm{op}} \le C\sqrt{1 + \frac{p}{n}} \le C\left(1 + \sqrt{\frac{p}{n}}\right).$$

Using Gaussian comparison inequalities one can even prove the above result with $C = 1$, *which is the best possible in light of the asymptotic result.*

Exercise 5.23. *Show that*

$$\|X\|_{\mathrm{op}} = \sup_{\operatorname{rank}(\Theta)=1, \|\Theta\|_F=1} \langle X, \Theta\rangle_F.$$

Exercise 5.24. *Show that*

(a) $\|AB\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}}\|B\|_F$.

(b) $\langle A, B\rangle_F \le \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}\|A\|_{\mathrm{op}}\|B\|_F$.

Theorem 5.25 (Chaining). *We have*

$$\mathbb{E}\sup_{t\in\mathbb{T}} X_t \le \mathbb{E}\sup_{t,t'\in T}(X_t - X_{t'}) \le 2\mathbb{E}\sup_{\substack{t,t'\in\mathbb{T},\\ \rho(t,t')\le\delta}}(X_t - X_{t'}) + 32\int_{\delta/4}^{D}\sqrt{\log N(u,\mathbb{T},\rho)}\,du$$

Proof. The first few steps of the proof are the same as those in the proof the crude discretization bound. We begin with the inequality

$$\mathbb{E}\sup_{t,t'\in T}(X_t - X_{t'}) \le 2\mathbb{E}\sup_{\substack{t,t'\in\mathbb{T},\\ \rho(t,t')\le\delta}}(X_t - X_{t'}) + 2\mathbb{E}\max_j |X_{t_j} - X_{t_1}|.$$

We will bound the second term more carefully this time. □

Example 5.26. *Going back to Example 5.21, if we use chaining, the resulting bound is*

$$\mathbb{E}\sup_{f\in\mathcal{F}} X_f \le 2\delta\sqrt{n} + 32\int_{\delta/4}^{D}\sqrt{\log C_{B,\nu} + 2(\nu-1)\log\left(\frac{1}{u}\right)}\,du.$$

The integral above can be upper-bounded by a constant multiple of $\int_0^D \sqrt{\log(\frac{1}{u})}\,du$*, which is finite. Hence, choosing* $\delta = \frac{1}{\sqrt{n}}$*, we get*

$$\mathbb{E}\sup_{f\in\mathcal{F}} X_f = O_{B,\nu}(1),$$

whence follows the optimal bound

$$\mathbb{E}_\eta \sup_{f\in\mathcal{F}} \left|\frac{1}{n}\sum_{i=1}^{n}\eta_i f(\xi_i)\right| = O_{B,\nu}\left(\frac{1}{\sqrt{n}}\right).$$

Exercise 5.27. *Show that* $\int_0^D \sqrt{\log(\frac{1}{u})}\,du < \infty$.

Remark 5.2. *Although chaining is able to remove extra* $\log n$ *factors in many problems, there are situations where it does not produce optimal bounds. Michel Talagrand's* generic chaining *method has the final say—it produces optimal upper bounds.*

We end this section by collecting in form of a theorem the results we have proved so far about a uniformly bounded function class of finite VC-dimension.

Theorem 5.28 (Glivenko-Cantelli for uniformly bounded function classes of finite VC-dimension). *Suppose* $\mathcal{F}$ *is a function class uniformly bounded by* B *and with finite VC-dimension* ν*. Then, with probability at least* $1 - e^{-\frac{n\delta^2}{2B^2}}$*, one has*

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \le O_{B,\nu}\left(\frac{1}{\sqrt{n}}\right) + \delta.$$

This is much stronger than just saying that $\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \overset{a.s.}{\to} 0$, which follows from the above Theorem by the Borel-Cantelli lemma. Note that if we choose $\delta^2 = \frac{2kB^2}{n}$, then, with probability at least $1 - e^{-k}$, we have

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} = O_{B,\nu}\left(\sqrt{\frac{k}{n}}\right),$$

which implies that

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} = O_P\left(\frac{1}{\sqrt{n}}\right),$$

which is in fact the optimal rate.

References

Billingsley, P. (2013). *Convergence of probability measures*. John Wiley & Sons.

Karatzas, I. and Shreve, S. (2012). *Brownian motion and stochastic calculus*, volume 113. Springer Science & Business Media.

Kumaresan, S. (2005). *Topology of metric spaces*. Alpha Science International Limited.

Munkres, J. R. (1975). *Topology: a first course*, volume 23. Prentice-Hall Englewood Cliffs, NJ.